

Market Integration, Educational Attainment, and the Effects of Displacement on Host Populations

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Abstract

Most forcibly displaced people are displaced within their own country and hosted in low- and middle-income countries, yet how their arrival affects host communities remains poorly understood, particularly in rural areas. This paper estimates the effects of conflict-induced internal displacement on hosts in Ethiopia between 2018 and 2022, combining household panel data with an enclave shift-share instrument that interacts the origin composition of a host area's existing displaced population with subsequent conflict shocks in those origin areas. Effects differ sharply across destinations. In rural areas, displacement raises local food prices by about 3 percent, leaves food expenditure unchanged once deflated, and worsens food security, while pulling hosts into household agricultural work and shifting production from cash crops towards staples — the pattern of a demand shock on food markets that local supply cannot accommodate. In urban areas, food prices fall, deflated food expenditure rises by about 17 percent, food security improves, and hosts move out of agriculture into wage employment. Splitting the sample on market integration reproduces the price and consumption divide without using the rural–urban classification, indicating that connectedness to outside markets, rather than the rural label itself, governs whether displacement is transmitted into prices. Food aid restrains prices where it reaches, but its coverage erodes roughly three times faster in rural areas as inflows grow. The educational composition of the workforce then determines which hosts can move between sectors in response.

JEL Classification: O15, J61, R23, Q11, O18

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1 Introduction

The number of people forcibly displaced worldwide doubled between 2014 and 2024, reaching 123.2 million by the end of the period ([UNHCR, 2024](#)). Yet policy discourse and media coverage have focused disproportionately on displacement movements toward high-income countries, even though low- and middle-income countries hosted 76% of the world’s displaced people by 2024. Public attention has also centered primarily on refugees, who cross international borders, while internally displaced people (IDPs)—those displaced within their own country’s borders—account for almost 60% of displacement worldwide ([UNHCR, 2024](#)). The result is a central gap in our understanding of displacement: comparatively little is known about the most prevalent form of displacement, particularly in rural areas of low- and middle-income countries. Understanding how internal displacement affects host communities is therefore central to designing policy responses that support displaced households while accounting for the communities that receive them.

This paper studies how internal displacement affects host communities in Ethiopia, where conflict-driven displacement rose sharply between 2018 and 2022. Using household panel data and a shift-share instrumental variable strategy, I estimate the effects of IDP arrivals on food prices, consumption, food security, and host labor supply in rural and urban destinations. The results reveal starkly different effects across the two settings: in rural areas, displacement raises food prices, leaves real food consumption unchanged, and pulls workers into agriculture; in urban areas, prices fall, food consumption and food security improve, and workers shift out of agriculture and into wage employment.

A simple framework in which displacement acts simultaneously on consumer goods markets and on labor markets helps interpret these results, with the balance between the two depending on the capacity of the local economy to adjust. In consumer goods markets, displacement operates as a demand shock. The arrival of displaced households raises local demand for consumption goods, especially food, and whether this translates into higher prices and lower real consumption depends on the ability of local supply to respond. Market integration and humanitarian assistance are therefore central: well integrated markets can absorb higher demand through trade, and food aid can relax pressure on local food markets directly, while poorly served areas transmit the demand shock into prices.

In labor markets, displacement operates as a labor supply shock. Its effect on hosts depends on their complementarity with displaced workers ([Borjas and Monras, 2017](#)). Displaced workers often experience occupational downgrading and enter low-skilled or informal jobs regardless of prior qualifications ([Lebow, 2024](#)). Hosts with low complementarity to displaced workers may face lower wages or reduced employment, while more educated, higher-skill, or more mobile hosts may benefit from complementarities with the displaced population. The two shocks are not independent, however: where the demand shock raises local food prices, it also raises the returns to producing food, so that the labor market response reflects the goods market outcome as well as the arrival of

workers.

Section 3 formalizes this logic in a stylized general equilibrium model of two locations, each producing food and a non-traded local service. The rural food market is modeled as being in autarky, while the urban food market is a small open economy that trades food at a fixed outside price; the service good is non-traded in both locations. Workers sort into the two sectors through Roy-style occupational selection on comparative advantage, in the tradition of [Lagakos and Waugh \(2013\)](#), adapted here to derive comparative statics for how prices and the sectoral allocation and hours of labor respond to the arrival of displaced people. The model shows that these two dimensions along which rural and urban destinations differ—market integration and the distribution of worker skill—are sufficient to generate the divergent price and labor-market responses documented below.

This framework helps reconcile three broad findings in the existing literature. One strand examines consumption, documenting increases in food and livestock prices alongside greater food insecurity among hosts, primarily in rural or semi-urban Sub-Saharan Africa ([Alix-Garcia and Saah, 2010](#); [George and Adelaja, 2022](#)). A second set of studies turns to labor markets and documents negative effects, particularly for low-skilled or informally employed workers ([Borjas and Monras, 2017](#); [Kleemans and Magruder, 2018](#); [Morales, 2018](#); [Aksu et al., 2022](#); [Ceritoglu et al., 2017](#)). A third finds null or positive labor market effects for hosts, often concentrated among higher-skill groups ([Card, 1990](#); [Alix-Garcia and Saah, 2010](#); [Fallah et al., 2019](#); [Winton, 2025](#)). Rather than viewing these results as contradictory, this paper shows that they can arise from the relative strength of consumer demand shocks, labor supply shocks, and local adjustment mechanisms, and that which of the three dominates depends on the setting in which displacement occurs.

The empirical setting is well suited to studying these mechanisms. Ethiopia experienced a sharp rise in conflict and displacement between 2018 and 2022, driven both by an increase in ethnic clashes throughout the country and by the Tigray war. To address the concern that IDPs settle where local conditions are favorable, I instrument a host area’s exposure to displacement with an enclave instrument in the tradition of the immigration literature: the composition of the displaced population already present around a host community, by woreda of origin, is combined with subsequent conflict shocks in those origin woredas. Identification relies on the assumption that conflict at IDPs’ places of origin is orthogonal to conditions at destination, conditional on destination conflict and fixed effects. Because a single origin shock reaches every destination hosting people from that woreda, regressions are estimated on a shift-level dataset, which allows standard errors to be clustered at the place of origin following [Borusyak et al. \(2022\)](#). Household and time fixed effects are included throughout.

The results show that displacement affects rural and urban host communities through different dominant mechanisms. In rural areas, food prices rise by about 3 percent relative to the national average when scaled to the average inflow, and real food expenditure is statistically indistinguishable from zero, so that nominal gains are absorbed by inflation rather than converted into additional consumption; food security deteriorates on suggestive measures. Labor markets respond by pulling

workers into agriculture, with hours in household agricultural work rising by roughly a quarter, while production itself is redirected from cash crops towards staples. In urban areas, by contrast, food prices fall by 2 to 3 percent, food expenditure deflated by local prices rises by about 17 percent, food security improves, and workers move out of agriculture and into wage employment.

Mechanism results point to the importance of food aid, market integration, and workforce composition. Food aid appears to contain inflationary pressure in both rural and urban areas, but its coverage erodes more quickly in rural areas following new displacement: the share of IDPs relying on food aid as their main source of food falls roughly three times as fast per thousand arrivals in rural areas as in urban ones. Assistance alone, however, cannot account for the divergence. Prices rise in rural areas precisely for those food items that humanitarian packages do not cover, while the same items remain contained in urban areas, which points to a further difference in how the two settings obtain food from outside. Market integration is that difference. Splitting the sample on measures of market access and integration reproduces the rural–urban divide in price and consumption outcomes without reference to the rural–urban classification itself, and market integration also explains part of the pull into agriculture.

Differences in labor market effects appear instead to reflect the composition of the host workforce. In rural areas, only 14 percent of workers completed primary school, compared with 59 percent in urban areas. Taking primary school completion as a proxy for skill and for mobility across sectors, rural areas hold a larger share of workers with fewer margins on which to respond. In rural settings, the pull into household agriculture is common to both education groups, but only more educated workers also increase hours in non-agricultural self-employment. In urban areas, workers at both education levels shift hours from household agriculture into wage employment; what separates them is that more educated workers replace the hours they lose in full, while less educated workers end up working about 4 percent fewer hours overall.

Taken together, the findings suggest that the effects of displacement depend on how consumer demand shocks and labor supply shocks interact with local adjustment capacity. In rural areas, weak market integration and declining aid coverage allow the demand shock to dominate in consumer goods markets, raising prices and leaving real food consumption unchanged. At the same time, a largely low-skilled host workforce has few avenues out of agriculture, so the labor response is channeled into food production rather than into other sectors. In urban areas, stronger market integration and more effective aid coverage absorb the increase in consumer demand, while a more educated workforce is better positioned to complete the shift into wage employment without losing hours.

Contribution to the literature. This paper contributes first to the literature on displacement and host communities, where the studies cited above appear to disagree about whether hosts gain or lose. Part of that disagreement, I argue, reflects a division of labor across strands rather than a genuine contradiction. The work documenting rising prices and worsening food insecurity

is concentrated in rural Sub-Saharan Africa, while the work on labor market effects is drawn largely from urban settings, where average effects are typically small and the interesting variation is distributional, running between low- and high-skill hosts. The two strands therefore examine different margins in different places, and neither is positioned to say whether the margin it studies behaves differently elsewhere. By estimating effects on prices, consumption, and labor supply together, in both rural and urban destinations, within one country, one survey, and one identification strategy, this paper makes the comparison the literature has had to make across papers. This is particularly valuable for rural areas, where few studies examine the labor market effects of immigration or displacement at all. Because the treatment is the flow of arrivals over the preceding year, these estimates describe short-run adjustment, before local supply, investment, and migration have had time to respond, and a longer horizon can look quite different (Morales, 2018; Imbert and Ulyssea, 2026).

Second, the paper contributes to the literature on structural transformation. Existing work has documented the frictions that prevent migration to urban centers (Bryan et al., 2014; Bryan and Morten, 2019; Baseler, 2023), the insurance benefits of rural out-migration (Lagakos et al., 2023), and the role of infrastructure in drawing workers out of agriculture (Asher and Novosad, 2020; Brooks and Donovan, 2020). This paper shows that displacement can also induce sectoral reallocation among host workers, but that the direction and distribution of this reallocation depend on market integration and worker skill. In less-integrated rural areas, movement out of agriculture is limited to more educated workers and occurs mainly through non-agricultural self-employment; in urban areas, workers shift more broadly from household agriculture into wage employment.

Outline. The remainder of the paper is organized as follows. Section 2 describes the study setting. Section 3 develops a stylized model of market integration and occupational selection that organizes the empirical results that follow. Section 4 presents the data and the construction of the analysis samples, and Section 5 the empirical strategy. Section 6 reports the main results, and Section 7 examines mechanisms. Section 8 concludes.

2 Context

2.1 Conflict and Displacement in Ethiopia, 2018–2022

After two decades of relative stability and rapid economic growth, Ethiopia saw conflict escalate again from 2016 in the form of inter-communal violence. By 2018 the country recorded the highest number of new conflict-related internal displacements ever registered anywhere, at 2.9 million over the year, four times the figure of the year before (IDMC, 2019).

This escalation coincided with a period of significant political change. Abiy Ahmed became prime minister in 2018, ended the state of emergency, released political prisoners, and brokered a peace agreement with Eritrea. He was credited with curbing the use of excessive force against protesters

and with opening the country to cooperation with humanitarian agencies. He also acknowledged the existence of conflict-induced internal displacement in Ethiopia, a reality the government had until then denied ([IDMC, 2019](#)). That acknowledgement matters for what follows, since it is what made systematic enumeration of displaced populations possible in the first place.

Most of the displacement that followed originated along the borders between the Oromia region and three of its neighbors: the Southern Nations, Nationalities and Peoples' (SNNP) region, Benishangul Gumuz, and the Somali region. Urban centers were affected as well, including Addis Ababa and Jijiga, the capital of the Somali region.

The single largest episode was the inter-communal violence between the Guji and Gedeo ethnic groups along the Oromia–SNNP border, which displaced some 700,000 people between April and June 2018 into overcrowded camps where humanitarian agencies struggled to meet basic needs ([OCHA, 2018c](#)). Because these agencies had no prior presence in the area, the government worked with them to establish a large-scale response from scratch. On the Oromia–Benishangul Gumuz border, ethnic violence following the killing of three police officers produced roughly 62,000 new displacements between October and December 2018, where needs were acute but humanitarian access was constrained by insecurity ([OCHA, 2018b](#)). Violence also broke out in the two urban centers: inter-ethnic clashes in Jijiga displaced 140,000 people within a few days in August 2018 ([OCHA, 2018a](#)), while violence surrounding the return of Oromo Liberation Front fighters from Eritrea displaced 15,000 people in Addis Ababa ([The New York Times, 2018](#)).

By the end of May 2019 the government had mounted a large-scale operation to return 1.9 million IDPs to their areas of origin. The operation was widely criticized for proceeding before the conditions for durable return—security and assistance at origin—had been met, and it was in part unsuccessful: nearly 90 percent of those returned to the Kamashi zone of Benishangul Gumuz are reported to have displaced again to the West Wollega zone of Oromia, citing insecurity and the absence of assistance ([OCHA, 2019b](#)).

The crisis took a new turn in 2020 with the outbreak of war in Tigray, in the north of the country. Tensions had been building since the purge of Tigrayan officials from the federal government, and as relations between Addis Ababa and Mekelle deteriorated, Tigrayan troops staged a mutiny in November 2020 and took command of the regional capital. The federal government responded with a full offensive, allied with paramilitary forces from the Amhara and Afar regions and with Eritrean military forces, while the newly established Tigray Defence Forces fought alongside the Tigray Special Forces and locally recruited militias. The war continued until a decisive federal advance forced Tigrayan forces to sue for peace in November 2022. The human toll was severe, with violations of international human rights and humanitarian law by all parties leaving tens of thousands killed and hundreds of thousands displaced ([International Crisis Group, 2026](#)). As explained in Section 4, the analysis sample does not cover the Tigray region itself, but the war had consequences for conflict and displacement in the neighboring Amhara and Afar regions, which the sample does cover.

By the end of 2021, cumulative new displacements across all conflicts in Ethiopia reached 5,142,000, once again the highest figure ever recorded for a single country ([IDMC, 2022](#)).

Displacement patterns. Where displaced people go is not arbitrary. Movement in Ethiopia generally follows routes determined by ethnic and socio-cultural affinity, with most IDPs fleeing to neighboring areas where they belong ethnically or share socio-cultural ties with the resident population ([REF, 2023](#)). The role of personal connections is more specific still: among a sample of drought-displaced people in the Somali region in January 2021, 69 percent were living with hosts belonging to their own close circle ([UNDP, 2022](#)). Displacement is thus channeled by social and family ties rather than distributed at random across feasible destinations.

The persistence of these channels is visible in the return episode described above. When returns to the Kamashi zone failed, the overwhelming majority of those affected displaced again to West Wollega, the destination of the original spell rather than a new one, which indicates that the same ties govern where people go on each occasion. This persistence is what the empirical strategy exploits. If the composition of a host area’s existing displaced population by area of origin reflects durable ties between the two places, then it also predicts where the next wave from those origins will settle, and the identifying variation can be built from shocks at origin rather than from conditions at destination. Section 5 develops this into the enclave instrument used throughout the analysis.

2.2 Living Conditions of IDPs and Humanitarian Assistance

Living conditions. By the end of 2019, 82 percent of Ethiopia’s IDPs had been displaced by conflict, with climate-related shocks such as drought and flooding accounting for most of the remainder ([OCHA, 2019c](#)). Their most pressing needs were access to livelihoods, restoration of land and property, and an adequate standard of living: shortages of food and water had produced widespread malnutrition, and low healthcare access left many vulnerable to disease.

The DTM assessment of late 2019 gives a more detailed picture, and one directly relevant to the analysis below. Thirty-seven percent of IDPs were living within host communities or with families. Of these, 86 percent reported access to livelihoods but only 12 percent had access to income, and a majority—60 percent—relied on food assistance as their main source of food. A further 33 percent lived in spontaneous settlements, where access to livelihoods was higher at 96 percent but access to income was the same 12 percent, and where reliance on food assistance was near-universal at 96 percent. Conditions in these settlements were poor: 4 percent of sites had access to electricity, 83 percent showed evidence of open defecation, and 70 percent had access to a health facility. The remaining IDPs were distributed across collective centers in repurposed buildings such as schools (9 percent), dispersed self-organized arrangements in rural areas (5 percent), planned camps (5 percent), and transit centers (2 percent).

Two features of this picture bear on the analysis. First, the great majority of IDPs settled among host populations rather than in planned camps, so that hosts and displaced people shared local

markets for food and labor. Second, the combination of high reported access to livelihoods with very low access to income indicates that most IDPs could work but earned little, which is the configuration under which their arrival registers on local markets primarily as an increase in demand for food and in the supply of labor.

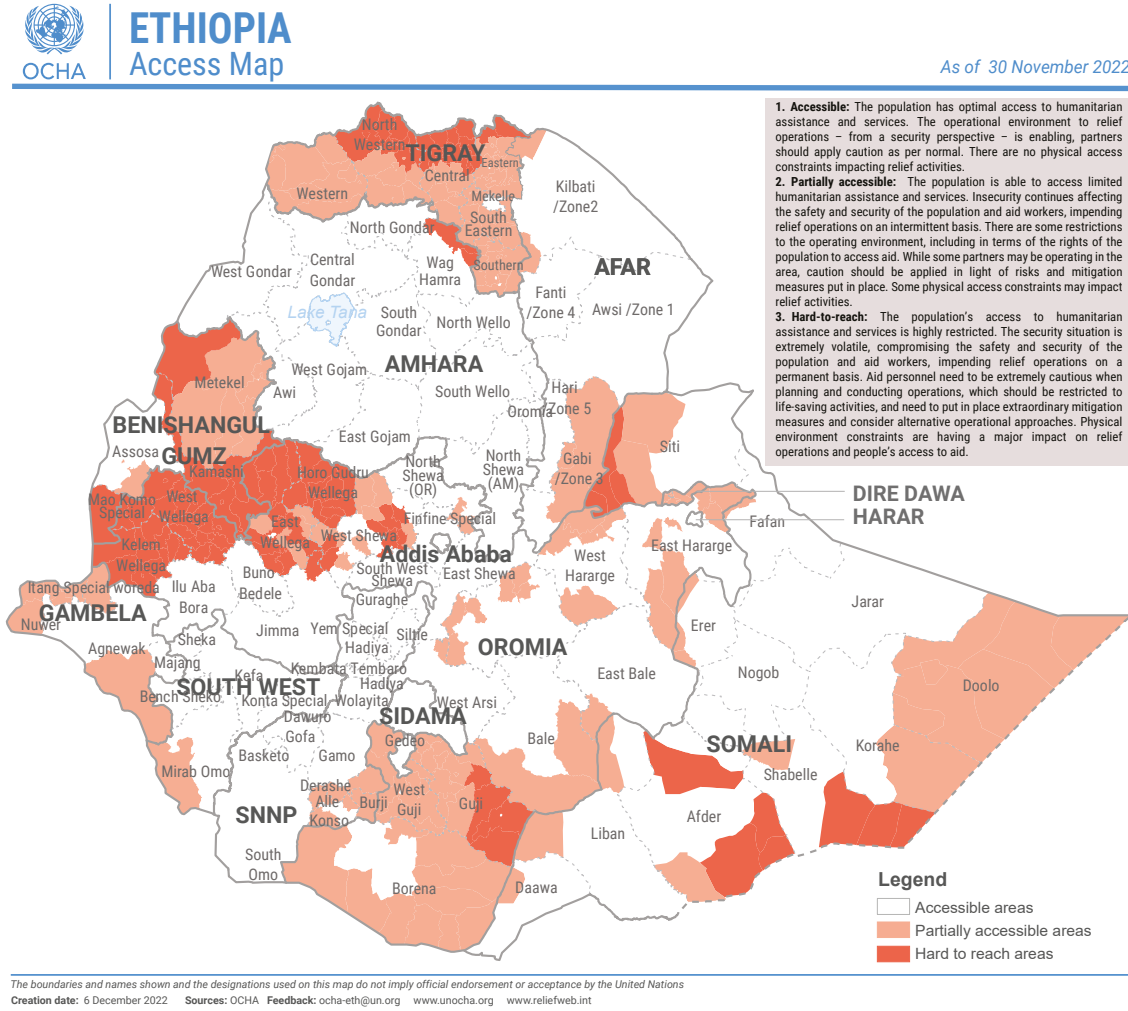
Humanitarian assistance. The response to conflict, displacement, and drought in Ethiopia was a large-scale effort involving the federal government, international agencies, and local organizations: six government entities, five UN agencies, 52 international NGOs, and 27 national NGOs, coordinated by the United Nations Office for the Coordination of Humanitarian Affairs (OCHA). The response is organized into clusters, each covering a thematic area and comprising the institutions active within it. Nine clusters operated in Ethiopia over this period, among them the Food cluster, led by the World Food Programme ([OCHA, 2022](#)).

The scale of assistance required grew substantially between 2018 and 2022, largely on account of the Tigray war. At the start of 2018 the Food cluster targeted five to seven million people; by the end of 2022 that figure had reached thirteen million ([OCHA, 2019a](#); [WFP, 2022](#)).

Delivering that assistance was constrained in ways that matter for the mechanisms examined in Section 7. Some constraints were logistical and security-related: northern Ethiopia and several other regions were inaccessible for extended periods, and the north was blocked entirely from August 2022 until the peace agreement in November, when deliveries resumed and reached 4.6 million of the thirteen million people targeted ([WFP, 2022](#)). Access problems were not confined to the north or to the final months of the period. By July 2022, areas of Tigray, Oromia, and Amhara were already inaccessible or only partly accessible to relief operations, as Figure 1 shows. Funding was a binding constraint as well: shortfalls led the Food cluster to reduce the number of people targeted in some regions, to distribute an incomplete food basket, and to apply ration cuts ([WFP, 2022](#)).

These constraints operated on top of the ordinary difficulty of reaching remote areas, and they are the reason assistance cannot be assumed to arrive wherever displaced people do. Whether food aid keeps pace with arrivals, and whether it does so differently in well-connected and poorly connected areas, is an empirical question that Section 7 takes up directly.

Figure 1: Limited access for humanitarian operations in November 2022 (Source: OCHA)



3 Conceptual Framework

This section presents a stylized model of how the same displacement shock can produce different price and labor-market responses in rural and urban host economies. The locations differ along two dimensions observed in the data: rural food markets are less integrated with outside markets, and rural workers are concentrated at lower skill levels. The model shows that these differences are sufficient to generate higher food prices and agricultural labor in rural areas, but stable food prices and a shift toward non-agricultural work in urban areas.

3.1 Environment

There are two locations, $e \in \{R, U\}$, and two locally produced goods. The food good, F , is potentially tradable, while the non-agricultural good, N , is a non-traded local service. The latter

includes retail, transport, construction, repairs, restaurants, and other activities that must be supplied locally. The rural food market is in autarky. The urban location is a small open economy in food and can import at the delivered outside price p_F^* . Local services are non-traded in both locations.

Individuals consume food and services and supply hours of work. Preferences are

$$U(c_F, c_N, h) = C(c_F, c_N) - v(h), \quad C(c_F, c_N) = \left(\frac{c_F}{\alpha}\right)^\alpha \left(\frac{c_N}{1-\alpha}\right)^{1-\alpha}, \quad v(h) = \chi \frac{h^{1+1/\varepsilon}}{1+1/\varepsilon}. \quad (1)$$

The parameter $\alpha \in (0, 1)$ is the expenditure share on food, $\chi > 0$ scales the disutility of work, and $\varepsilon > 0$ is the elasticity of hours with respect to the skill-adjusted real wage. The consumption aggregator $C(c_F, c_N)$ converts a bundle of food and services into units of composite consumption. Its price index is

$$P_e = p_{Fe}^\alpha p_{Ne}^{1-\alpha}. \quad (2)$$

Cost minimization implies that consumers allocate shares α and $1 - \alpha$ of expenditure to food and services, respectively.

Let E_e^H denote aggregate nominal host expenditure, held fixed when tracing the first-round effect of displacement, and write $\theta_F = \alpha$ and $\theta_N = 1 - \alpha$. Cobb–Douglas preferences imply host demand $Q_{se}^H = \theta_s E_e^H / p_{se}$. Each displaced household brings outside purchasing power $b_e > 0$ and allocates the same expenditure shares to the two goods. Total demand is therefore

$$Q_{se}^d(p_{se}, D) = \frac{\theta_s (E_e^H + b_e D)}{p_{se}}. \quad (3)$$

Define the direct proportional shift in demand, holding the price fixed, as

$$\kappa_{se} \equiv \frac{1}{Q_{se}^d} \frac{\partial Q_{se}^d}{\partial D} = \frac{b_e}{E_e^H + b_e D} \equiv \kappa_e > 0. \quad (4)$$

Thus $\kappa_{Fe} = \kappa_{Ne} = \kappa_e$: Cobb–Douglas expenditure shares make the displacement-induced proportional demand shift identical across the two sectors.¹ Conditional on nominal expenditure, total demand is also unit elastic with respect to its own price. Imposing the same elasticity in both sectors shuts down differences in demand elasticities as an explanation for the relative-price result; holding other forces fixed, less price-elastic food demand would reinforce that result. This partial-equilibrium formulation abstracts from feedback from equilibrium wages and profits into the host demand schedule.

On the production side, let $S_{se}(p_{se}, D)$ denote the short-run supply of good $s \in \{F, N\}$. The supply schedule summarizes firms' labor demand and workers' equilibrium labor supply. It is increasing in

¹The equality of the sectoral demand shifts is therefore not an additional assumption in Proposition 1. It follows from common Cobb–Douglas expenditure shares and from holding host nominal expenditure fixed in the first-round comparative static.

the output price. Define its own-price elasticity and its direct response to the arrival of displaced workers as

$$\eta_{se} \equiv \frac{p_{se}}{S_{se}} \frac{\partial S_{se}}{\partial p_{se}} > 0, \quad \lambda_{se} \equiv \frac{1}{S_{se}} \frac{\partial S_{se}}{\partial D} \geq 0. \quad (5)$$

The term λ_{se} allows displaced workers to expand local supply. Since the arrivals are predominantly low-skilled, this direct supply response may be larger in food production than in services. The goods-price comparative statics depend on the demand shift net of this direct supply response,

$$\delta_{se} \equiv \kappa_{se} - \lambda_{se}. \quad (6)$$

3.2 Goods-market adjustment

Both goods must clear locally in the rural economy:

$$S_{FR}(p_{FR}, D) = Q_{FR}^d(p_{FR}, D), \quad S_{NR}(p_{NR}, D) = Q_{NR}^d(p_{NR}, D). \quad (7)$$

For sector s , differentiating its market-clearing condition with respect to D gives

$$\frac{\partial S_{sR}}{\partial p_{sR}} \frac{dp_{sR}}{dD} + \frac{\partial S_{sR}}{\partial D} = \frac{\partial Q_{sR}^d}{\partial p_{sR}} \frac{dp_{sR}}{dD} + \frac{\partial Q_{sR}^d}{\partial D}, \quad (8)$$

Because host and displaced demand are both unit elastic, total demand is also unit elastic. Dividing equation (8) by equilibrium quantity and applying the definitions in equations (4)–(6) yields

$$\eta_{sR} \frac{d \log p_{sR}}{dD} + \lambda_{sR} = \kappa_{sR} - \frac{d \log p_{sR}}{dD}. \quad (9)$$

Rearranging gives the rural price comparative static:

$$\frac{d \log p_{sR}}{dD} = \frac{\delta_{sR}}{1 + \eta_{sR}}, \quad s \in \{F, N\}. \quad (10)$$

The one in the denominator is the unit own-price elasticity implied by Cobb–Douglas demand. It follows that

$$\frac{d}{dD} \log \left(\frac{p_{FR}}{p_{NR}} \right) = \frac{\delta_{FR}}{1 + \eta_{FR}} - \frac{\delta_{NR}}{1 + \eta_{NR}}. \quad (11)$$

In the urban economy, the outside market pins down the food price and imports clear excess food demand:

$$p_{FU} = p_F^*, \quad M_U = Q_{FU}^d - S_{FU}. \quad (12)$$

Services must still clear locally. Consequently,

$$\frac{d \log p_{FU}}{dD} = 0, \quad \frac{d \log p_{NU}}{dD} = \frac{\delta_{NU}}{1 + \eta_{NU}}. \quad (13)$$

Proposition 1: Product prices. Suppose that (i) rural food supply is less price-elastic than rural service supply, $\eta_{FR} < \eta_{NR}$; (ii) net demand pressure is positive in rural and urban services, $\kappa_R > \lambda_{NR}$ and $\kappa_U > \lambda_{NU}$; and (iii) the excess direct supply response in rural food is bounded by

$$\lambda_{FR} - \lambda_{NR} < \frac{(\kappa_R - \lambda_{NR})(\eta_{NR} - \eta_{FR})}{1 + \eta_{NR}}. \quad (14)$$

Then displacement raises the relative price of food in R and lowers it in U :

$$\frac{d}{dD} \log \left(\frac{p_{FR}}{p_{NR}} \right) > 0, \quad \frac{d}{dD} \log \left(\frac{p_{FU}}{p_{NU}} \right) < 0. \quad (15)$$

Condition (iii) allows the predominantly low-skilled inflow to expand rural food supply more than service supply. This force dampens the food-price response. The condition requires it not to outweigh the greater short-run rigidity of food supply. In the very short run, land is fixed and planting decisions have often already been made, so η_{FR} may be close to zero. Informal trade and services can scale up with little fixed investment, making η_{NR} comparatively large. This elasticity gap enlarges the right-hand side of condition (iii) and allows a sizable difference between λ_{FR} and λ_{NR} . If the direct supply responses are equal, $\lambda_{FR} = \lambda_{NR}$, condition (iii) holds automatically.

Condition (iii) also guarantees positive net demand pressure in the rural food market. Because $\eta_{FR} < \eta_{NR}$, $0 < (\eta_{NR} - \eta_{FR})/(1 + \eta_{NR}) < 1$. Together with $\kappa_R > \lambda_{NR}$, condition (iii) therefore implies $\lambda_{FR} - \lambda_{NR} < \kappa_R - \lambda_{NR}$, or $\kappa_R > \lambda_{FR}$. Thus $\delta_{FR} = \kappa_R - \lambda_{FR} > 0$: holding prices fixed, the direct demand shift in rural food exceeds the direct supply shift.

To see the rural result, condition (iii) can be rearranged as

$$\frac{\kappa_R - \lambda_{FR}}{1 + \eta_{FR}} > \frac{\kappa_R - \lambda_{NR}}{1 + \eta_{NR}}, \quad (16)$$

which, by equation (11), is exactly the claimed increase in the rural relative price of food. In U , additional food demand is met by imports, while additional service demand must be met by local production.

3.3 Workers and sectoral adjustment

The labor-allocation block follows a Roy-style selection mechanism, as in [Lagakos and Waugh \(2013\)](#), in which heterogeneous workers sort between food and non-food production according to comparative advantage. The mechanism and object of analysis differ, however. [Lagakos and Waugh \(2013\)](#) use Stone–Geary subsistence demand and cross-country productivity differences to generate relative prices and selection, whereas the present model studies a short-run displacement shock interacting with local market integration and sectoral supply elasticities. Cobb–Douglas demand deliberately shuts down differences in conditional own-price demand elasticities between food and services, and the one-dimensional skill distribution varies across rural and urban locations.

Incumbent workers differ in a skill index z , distributed according to $G_e(z)$. A type- z worker supplies $a_F(z) = 1$ efficiency unit per hour in food and $a_N(z) = z$ efficiency units per hour in services. Thus, higher values of z represent greater productivity in non-agricultural work. The rural distribution G_R is concentrated at low values of z , whereas the urban distribution G_U places substantial mass at both low and high skill levels.

Competitive firms in sector s produce according to

$$Y_{se} = A_s L_{se}^{\beta_s}, \quad 0 < \beta_s < 1, \quad (17)$$

where L_{se} denotes efficiency units of labor. Decreasing returns to labor summarize short-run sectoral constraints, such as land in agriculture and premises or capital in services; these constraints are absorbed into A_s . Profit maximization implies

$$w_{se} = p_{se} A_s \beta_s L_{se}^{\beta_s - 1}, \quad (18)$$

where w_{se} is the wage per efficiency unit.

Conditional on working in sector s , a worker faces

$$P_e C = a_s(z) w_{se} h. \quad (19)$$

The first-order condition for hours is

$$\frac{a_s(z) w_{se}}{P_e} = v'(h) = \chi h^{1/\varepsilon}, \quad (20)$$

and therefore

$$h_s(z) = \left(\frac{a_s(z) w_{se}}{\chi P_e} \right)^\varepsilon. \quad (21)$$

Because indirect utility is increasing in the skill-adjusted wage, a worker chooses services if and only if

$$z w_{Ne} \geq w_{Fe} \iff z \geq \bar{z}_e \equiv \frac{w_{Fe}}{w_{Ne}}. \quad (22)$$

The share of incumbent workers in food production is $G_e(\bar{z}_e)$. Its response to displacement is

$$\frac{dG_e(\bar{z}_e)}{dD} = g_e(\bar{z}_e) \bar{z}_e \left[\frac{d \log w_{Fe}}{dD} - \frac{d \log w_{Ne}}{dD} \right]. \quad (23)$$

Thus the direction of reallocation is determined by relative wages, while its magnitude is governed by the density of workers at the cutoff.

To connect product prices to real wages, first hold the occupational cutoff locally fixed. Let $\mathcal{Z}_{Fe} = \{z : z < \bar{z}_e\}$ and $\mathcal{Z}_{Ne} = \{z : z \geq \bar{z}_e\}$ denote the sets of workers assigned to the two sectors. Efficiency units of labor equal the sum of each worker's hours multiplied by that worker's sectoral

productivity:

$$L_{se} = \int_{z \in \mathcal{Z}_{se}} a_s(z) h_s(z) dG_e(z). \quad (24)$$

Substituting equation (21) gives

$$L_{se} = B_{se} \left(\frac{w_{se}}{P_e} \right)^\varepsilon, \quad B_{se} \equiv \chi^{-\varepsilon} \int_{z \in \mathcal{Z}_{se}} a_s(z)^{1+\varepsilon} dG_e(z). \quad (25)$$

The exponent $1 + \varepsilon$ appears because productivity affects both the efficiency units contributed per hour and, through the real wage, the hours chosen. Holding the cutoff fixed makes B_{se} constant. Combining equations (25) and (18) then implies

$$\frac{d \log(w_{se}/P_e)}{dD} = \frac{1}{1 + \varepsilon(1 - \beta_s)} \frac{d \log(p_{se}/P_e)}{dD}. \quad (26)$$

Allowing workers at the cutoff to reallocate makes sectoral labor supply more elastic. Appendix A.1 shows that this reallocation attenuates the response of the relative wage w_{Fe}/w_{Ne} to the relative product price p_{Fe}/p_{Ne} without changing its direction. Equation (26) continues to describe each sector's real-wage response exactly when no workers lie at the cutoff and approximately when the density there is small. From equation (2),

$$d \log(p_{Fe}/P_e) = (1 - \alpha) d \log(p_{Fe}/p_{Ne}), \quad (27)$$

$$d \log(p_{Ne}/P_e) = -\alpha d \log(p_{Fe}/p_{Ne}). \quad (28)$$

These are accounting identities rather than additional comparative statics: with two goods there is only one independent relative price. Their role is to express the product-price change from Proposition 1 in units of the local consumption basket, because hours depend on w_{se}/P_e . In particular, a decline in p_{Ne}/P_e means that the service price falls relative to the local cost of living; it need not fall in nominal terms. For sector choice, the price index cancels from the relative wage w_{Fe}/w_{Ne} . At a fixed cutoff, Proposition 1 therefore raises the real food wage and lowers the real service wage in R , while doing the reverse in U . These individual real-wage signs continue to hold when the density at the cutoff is sufficiently small. More generally, even with a moving cutoff, the relative food wage rises in R and falls in U .

Proposition 2: Labor allocation. Under the conditions of Proposition 1, displacement raises the relative food wage in R and the relative service wage in U . Consequently,

$$\frac{d\bar{z}_R}{dD} > 0, \quad \frac{d\bar{z}_U}{dD} < 0. \quad (29)$$

The share of incumbent workers in food production is nondecreasing in R and nonincreasing in U . The rural change is strict if $g_R(\bar{z}_R) > 0$, and the urban change is strict if $g_U(\bar{z}_U) > 0$; either response is zero if no workers in that location lie at the cutoff. If $g_R(\bar{z}_R)$ is small because most

rural workers are already in agriculture, rural reallocation is limited. When no rural workers lie at the cutoff, or when the density there is sufficiently small, equation (21) implies that the increase in the real food wage raises hours among existing agricultural workers:

$$\frac{d \log h_F(z)}{dD} = \varepsilon \frac{d \log(w_{FR}/P_R)}{dD} > 0. \quad (30)$$

Agricultural labor can therefore expand without a quantitatively important contraction in rural non-agricultural work. In U , a more balanced skill distribution places greater mass near the cutoff, producing a larger shift from food production into non-agricultural work. The model does not sign the change in total hours of low-skilled urban workers.

3.4 Empirical implications

The model delivers four predictions. First, displacement raises food prices relative to service prices in poorly integrated rural markets. Second, food prices are insulated from local demand in integrated urban markets, while the price and output of local services increase. Third, rural agricultural labor expands mainly through higher hours among workers already in agriculture when few rural workers are close to the occupational cutoff. Fourth, urban workers shift toward non-agricultural work when the urban skill distribution places substantial mass near that cutoff. Market integration determines which sector absorbs the demand shock, and the local skill distribution determines how incumbent labor follows the resulting change in returns.

Perfect food-market integration pins down the urban food price but does not by itself make that price fall. A negative urban food-price response can arise from an outward supply shift, such as in-kind food assistance or lower delivery costs. This extension is not required for the model’s central comparison between price adjustment in rural markets and quantity adjustment in urban markets.

4 Data

The analysis brings together three data sources: the Ethiopian Socioeconomic Panel Survey (ESS), which supplies outcomes and controls; the Displacement Tracking Matrix (DTM), which supplies the measure of exposure to displacement; and the Armed Conflict Location and Event Data (ACLED) project, which supplies conflict measures used both as instruments and as controls. The unit of observation in the resulting datasets is a household or an individual from the host community—that is, as explained below, one already residing in the area when the updated Ethiopian Central Statistical Agency (CSA) 2018 pre-census cartographic database of enumeration areas was created.² The timing of the sampling frame relative to the exposure window is what allows these households to be read as hosts rather than as displaced people themselves. Households

²The precise date on which the pre-census cartographic exercise was completed is not known to me, but it must precede September 2018, when fieldwork for the Ethiopian Socioeconomic Panel Survey (ESS) and the 2018 Annual Agricultural Sample Survey began.

enter the ESS through a frame constructed in 2018, in all likelihood before September, and are first interviewed in June 2019, while the treatment used below is the flow of IDPs arriving in the surrounding area over the year preceding that interview. The displacement being measured therefore post-dates the enumeration of the sampled households in place, which makes it unlikely that they are themselves among the IDPs whose arrival constitutes the treatment. Because the ESS is a panel drawn once from that 2018 frame, this argument does not depend on which wave a household is observed in: the frame pre-dates the exposure window in both waves, so a sampled household has little chance of being one of the displaced households counted in its own treatment variable.

This reasoning defines the host community somewhat differently from what is customary. What the timing of the frame rules out is displacement into the area during the exposure window—the year preceding a household survey interview, or the eight months preceding a post-planting interview—rather than displacement at any point in the past. A household that arrived as an IDP before the 2018 frame was constructed, and was enumerated as a resident of the area, is accordingly treated here as a host. That case is not merely hypothetical, since the analysis sample is by construction restricted to areas already hosting IDPs in early to mid-2018. Throughout what follows, then, a host should be understood as any household not displaced within the exposure window, rather than as one with no history of displacement at all.

4.1 ESS Data

Outcome and control variables come from waves 4 and 5 of the Ethiopian Socioeconomic Panel Survey, produced by the Ethiopian Statistical Service with technical support from the World Bank’s Living Standards Measurement Study (LSMS), and used here at the individual level. Each wave bundles a household survey, a community survey, and three agricultural surveys covering the post-planting period, the post-harvest period, and livestock. These modules are fielded at different points in the agricultural calendar, so that a single wave spans close to a year: in wave 4, the post-planting and livestock surveys ran from September to October 2018, the post-harvest survey from February to March 2019, and the household and community surveys from June to August 2019, while in wave 5 the post-planting and livestock surveys ran from September 2021 to January 2022 and the post-harvest, household, and community surveys from April to May 2022. This staggering matters for the analysis below, since outcomes drawn from different modules are observed at different dates and are therefore matched to different rounds of displacement data.

The survey is representative nationally and within each of Ethiopia’s eleven regions, with the exception of Tigray, which is excluded from wave 5. It also draws an explicit distinction between urban and rural areas, a distinction the analysis relies on throughout. The CSA classifies an enumeration area (EA) as urban if it is a locality of at least 2,000 inhabitants, if it serves as an administrative capital at the regional, zonal, or woreda level, or if it is a locality of 1,000 or more inhabitants who are not primarily engaged in agriculture (OECD and PSI, 2020); all remaining localities are classified as rural. Sampling proceeds in two stratified stages, with EAs drawn from

the updated CSA 2018 pre-census cartographic database. Ten agricultural and two non-agricultural households were then randomly selected within each rural EA, against fifteen households selected without regard to economic activity within each urban EA. The household surveys also report the GPS coordinates of each EA, though these are displaced for confidentiality: the offset ranges from 0 to 2 km in urban areas and from 0 to 5 km in rural areas, and reaches 0 to 10 km for a randomly chosen 1% of EAs, which raises locational uncertainty at the cost of little additional noise overall.

Expenditure outcomes are constructed by the ESS at the annual level and cover total, food, and non-food household expenditures, where the latter two are sub-aggregates built from selected items and therefore do not mechanically sum to the total. Because the analysis compares expenditures across two survey years, I convert Ethiopian Birr values in year t , X_t^{ETB} , into constant 2021 international dollars, X^{USD} :

$$X^{USD} = \frac{X_t^{ETB} \times (CPI_{2021}/CPI_t)}{PPP_{2021}} \quad t \in \{2019, 2021\} \quad (31)$$

where the Ethiopian PPP conversion factor for private consumption in 2021, PPP_{2021} , is set to 13.4367609024048 following the World Bank International Comparison Program database.

Food price outcomes are built from the household and community modules and take three forms. Two are price deflators that average, item by item, the ratio of the local price to the national average, weighted by household expenditure shares; they differ only in the level at which local prices are averaged, the first at the woreda level and the second at the EA level. The third is a standardized price index. Appendix section A.2 sets out the construction of all three in detail. Prices for non-food items are unfortunately not collected in the ESS, so the price analysis is necessarily confined to food.

The ESS also carries a food security module, from which I construct index measures of household food security. These are imperfect, because the recall period attached to the questions changes between the two waves used here: every question in wave 4 asks about the seven days preceding the interview, while every question in wave 5 asks about the preceding year. Table ?? lists the questions asked in each wave alongside their recall windows. Five questions cover the same dimensions of food insecurity in both waves, while others appear in only one. I therefore construct two indices: a full index drawing on every available question, and a restricted index using only the five dimensions common to both waves. Both are built the same way. Each question is coded as a binary indicator, the indicators are averaged within household and wave, and the resulting index is standardized within wave so that it has mean zero and unit variance in each. The index is finally multiplied by -1 , so that higher values denote greater food security rather than greater food insecurity. Appendix section A.3 discusses more in detail the issues with these indices, whose results will be taken as suggestive as they conflate at best seasonal food security (and at worst some structural relationship between treatment intensity and the type of food security households experience) with treatment effects. The ranking of household in terms of percentiles in each waves for the measure is also used

in the analysis to alleviate some of the concerns related to the measures.

The labor module records both participation and intensity. A first indicator captures whether an individual took part in any economic activity at all; three further indicators identify participation, over the seven days preceding the interview, in agricultural work for one’s own household, in self-employment, and in permanent wage employment, with hours worked recorded alongside each. Two residual activities, temporary wage employment and unpaid apprenticeships, are too rare to support separate analysis and are not treated as categories of their own, but they do enter the construction of total hours spent in any income-generating activity (IGA) together with the three main categories.

The agricultural survey supplies a further set of outcomes: family hours worked on farm during the planting season, person-days and daily wages for hired labor over the same period, indicators for the cultivation of staple and cash crops, and the potential rental value of land per hectare. Since this module is fielded only in rural areas, it permits an unusually detailed account of how displacement affects agricultural production there, but it leaves the corresponding question for more urban areas out of reach. Finally, the ESS provides the individual- and household-level controls used in the regressions: household size, the age and gender of the individual, and indicators for ever having attended school and for being able to read and write.

4.2 DTM Data

Exposure to displacement is measured using Ethiopian data from the Displacement Tracking Matrix, an instrument of the United Nations’ International Organization for Migration that documents displaced populations in more than eighty countries. In Ethiopia, several site assessments per year record internal displacement figures alongside IDPs’ needs and the services available where they have settled. Coverage is national in principle, but the areas actually assessed are determined by expected IDP caseloads; within an assessed area, every site hosting twenty or more IDPs is surveyed. Each round proceeds in three stages: key informants at the Disaster Prevention and Preparedness Office are interviewed first to establish the picture at the zone level, informants at the woreda level are then consulted in order to plan the assessments, and the site assessments themselves finally collect detailed information through key informant interviews.

Conversations with the Data and Research Unit at IOM Ethiopia clarified how these site assessments are conducted in practice. A typical assessment draws on seven key informants, all of them IDPs living at the site, of whom at least three are women, one is a youth, and one is an older person, with these categories permitted to overlap. What emerges from these interviews is then triangulated against independent sources such as shelter counts, camp registration lists, and household-level data.

The first assessment dates from January 2018, after which rounds were conducted at regular intervals of roughly three to six per year through 2023, with a small number of further assessments

in 2024. The unit of observation is the IDP site, and the data record its location including GPS coordinates, the number of IDPs present, the woreda from which the majority of them originate, and the humanitarian support they receive.

4.3 ACLED Data

The third source is the Armed Conflict Location and Event Data project for Ethiopia, whose staff compile a database of conflict events by reviewing journalistic and media sources, supplemented by material from the Ethiopian diaspora abroad and from official government media agencies. Each observation is a conflict event, reported with its precise location and exact type. From these records I retain the number of conflict events, the number of fatalities, and the number of events involving civilian targeting, which enter the analysis either as instruments or as control variables.

4.4 Construction of Analysis Datasets

The analysis rests on two datasets built from these sources. The household analysis dataset takes its outcomes from the ESS household survey and its skeleton from waves 4 and 5 of that survey, conducted in 2019 and 2022; each observation is an individual from the host community, meaning one sampled in or before 2018. The agricultural analysis dataset serves to estimate the effects of displacement on agricultural production, and is built instead on waves 4 and 5 of the post-planting surveys, where the unit of observation is the household.

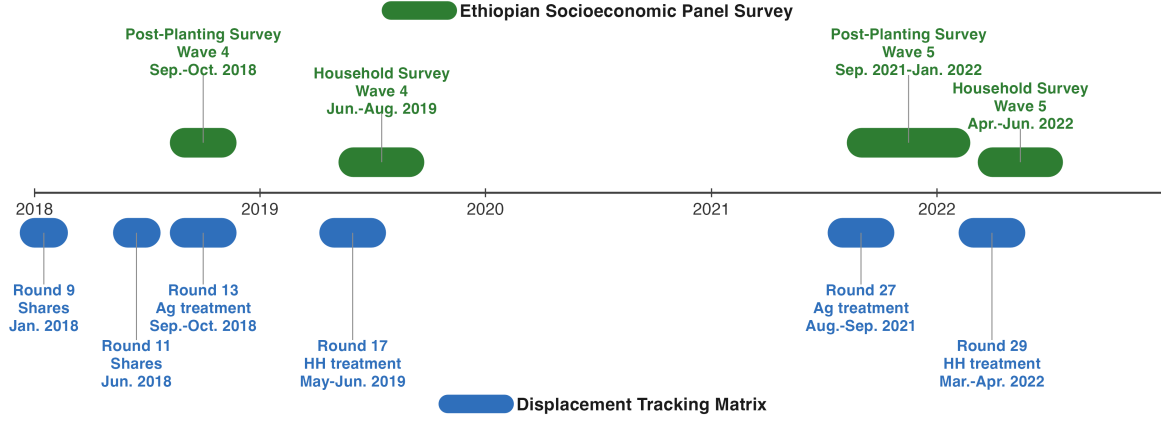
Onto each skeleton I merge the treatment variable, a household’s exposure to displacement, from the DTM. Because ESS modules are fielded at different times of year, the two datasets are matched to different assessment rounds: rounds 17 (May–June 2019) and 29 (March–April 2022) are merged with waves 4 and 5 of the household survey, while rounds 13 and 27 are merged with the corresponding post-planting surveys. Figure 2 places the ESS and DTM rounds used here on a common timeline. The merge itself is spatial: a circle of 20 km radius is drawn around the location of a host household’s EA, and the IDPs recorded at every DTM site falling inside that circle are summed.

Exposure enters the analysis as a flow rather than a stock. For outcomes from the household survey, the treatment is the one-year flow of IDPs arriving around a host household, measured in thousands. Since DTM rounds are not spaced exactly twelve months apart, the lagged round cannot be pinned to a fixed date: it is taken to be the most recent assessment conducted at least one year before the interview, so that the flow covers a span of at least, though rarely exactly, one year. For wave 4 observations this lagged round is generally round 11, so that the treatment is in practice the difference between the IDP counts of rounds 17 and 11, divided by one thousand. Formally, let $\mathcal{D}_{ht}^{20}$ denote the set of IDP sites d lying within 20 km of host household h ,

$$\mathcal{D}_{ht}^{20} = \{d : \text{dist}(h, d) \leq 20 \text{ km}\},$$

and let s_{dt}^0 and s_{dt}^{-1} denote, respectively, the latest DTM round preceding the ESS interview date t

Figure 2: Timeline for Ethiopian Socioeconomic Survey and Displacement Tracking Matrix data



Note: The timeline covers January 2018 through December 2022. Green bars above the timeline indicate Ethiopia Socioeconomic Survey fieldwork periods. Blue bars below the timeline indicate Displacement Tracking Matrix rounds used to construct displacement shares and treatment measures. Month ranges are shown inclusively.

and the latest round falling at least one year before it,

$$s_{dt}^0 = \max \{s : s \leq t\}, \quad s_{dt}^{-1} = \max \{s : s \leq t - 365\}.$$

Subtracting the IDP count at s_{dt}^{-1} from that at s_{dt}^0 and scaling by one thousand yields the treatment:

$$\Delta \text{IDP}_{ht}^{1yr, 1000} = \frac{1}{1000} \sum_{d \in \mathcal{D}_{ht}^{20}} \left(\text{IDP}_{ds_{dt}^0} - \text{IDP}_{ds_{dt}^{-1}} \right). \quad (32)$$

The same construction is not available for agricultural outcomes. The post-planting survey of wave 4 was fielded in September and October 2018, so a lagged round a full year earlier would fall before the DTM series begins with round 9 in January 2018. The constraint binds only for that wave—the wave 5 post-planting survey is late enough that a one-year lag would be feasible—but applying different windows across waves would make the treatment non-comparable between them, and I therefore shorten the window to eight months throughout, replacing the reference rounds with

$$u_{dt}^0 = \max \{u : u \leq t\}, \quad u_{dt}^{-1} = \max \{u : u \leq t - 244\},$$

so that

$$\Delta \text{IDP}_{ht}^{8mths, 1000} = \frac{1}{1000} \sum_{d \in \mathcal{D}_{ht}^{20}} \left(\text{IDP}_{du_{dt}^0} - \text{IDP}_{du_{dt}^{-1}} \right). \quad (33)$$

Sample restrictions. Four restrictions shape the final analysis samples. The first follows from the geographic coverage of the DTM: some areas went unassessed, for reasons ranging from insecurity to low expected caseloads, and rather than impute exposure for host households living in them, I drop them altogether. The second follows from the empirical strategy set out below, which uses the shares of IDPs surrounding host households before exposure to the treatment; these shares are measured at round 11 for regressions using household survey outcomes and in January 2018 for those using post-planting outcomes, so that only households already exposed to IDPs within 20 km of their EA centroid can remain in the sample. The third is the requirement of a balanced panel, which permits household fixed effects and forecloses attrition, and which restricts the sample to households observed at both wave 4 and wave 5 in the relevant survey. The fourth simply limits the sample to individuals aged 15 or older, the population whose labor market outcomes are of interest here.

Table 1 traces the effect of each restriction on sample size. All four cut deeply, and the number of unique households lost to the balanced panel requirement is perhaps larger than one would expect. Taken together, the restrictions mean the resulting samples should be read as representative of households that lived in areas assessed by the DTM, were already exposed to displacement in early to mid-2018, and were surveyed in both waves of the ESS. They form a balanced household-level panel, and can no longer be treated as representative of Ethiopia nationally or regionally.

Descriptive statistics. Figure 3 maps the EAs of the host households in the household analysis sample at wave 4 (2019), distinguishing urban from rural EAs, against the IDP sites recorded in the DTM. Table 2 summarizes their exposure to IDP flows. Over a year, host areas receive roughly 3,470 additional IDPs in rural areas and 5,390 in urban ones. Measured against the population living within 20 km of the EA centroid, the two are closer than these absolute figures suggest, at 2.71% of the surrounding population in rural areas and 2.29% in urban areas, since the larger urban inflows arrive in more populated areas. This relative measure is reported for interpretation only and plays no part in estimation. Exposure is somewhat lower in the agricultural sample, which contains fewer observations and shows smaller inflows in both absolute and relative terms. Alongside the treatment, the samples carry conflict variables built from ACLED: the numbers of conflict events, of events with fatalities, and of events involving civilian targeting occurring within 20 km of the EA centroid, each summed over the calendar year of the survey, 2019 for wave 4 and 2022 for wave 5.

Table 3 reports summary statistics for the household analysis sample, separately for rural and urban households. Expenditures are higher among urban households despite their smaller size. Food expenditures are the exception, higher in rural areas before local prices are accounted for but lower once deflated by them, which indicates that rural households spend more on food because the food they buy is more expensive. The urban advantage is starkest for non-food expenditures, which are more than twice their rural counterparts.

Table 1: Number of unique households and individuals with restrictions applied to analysis samples

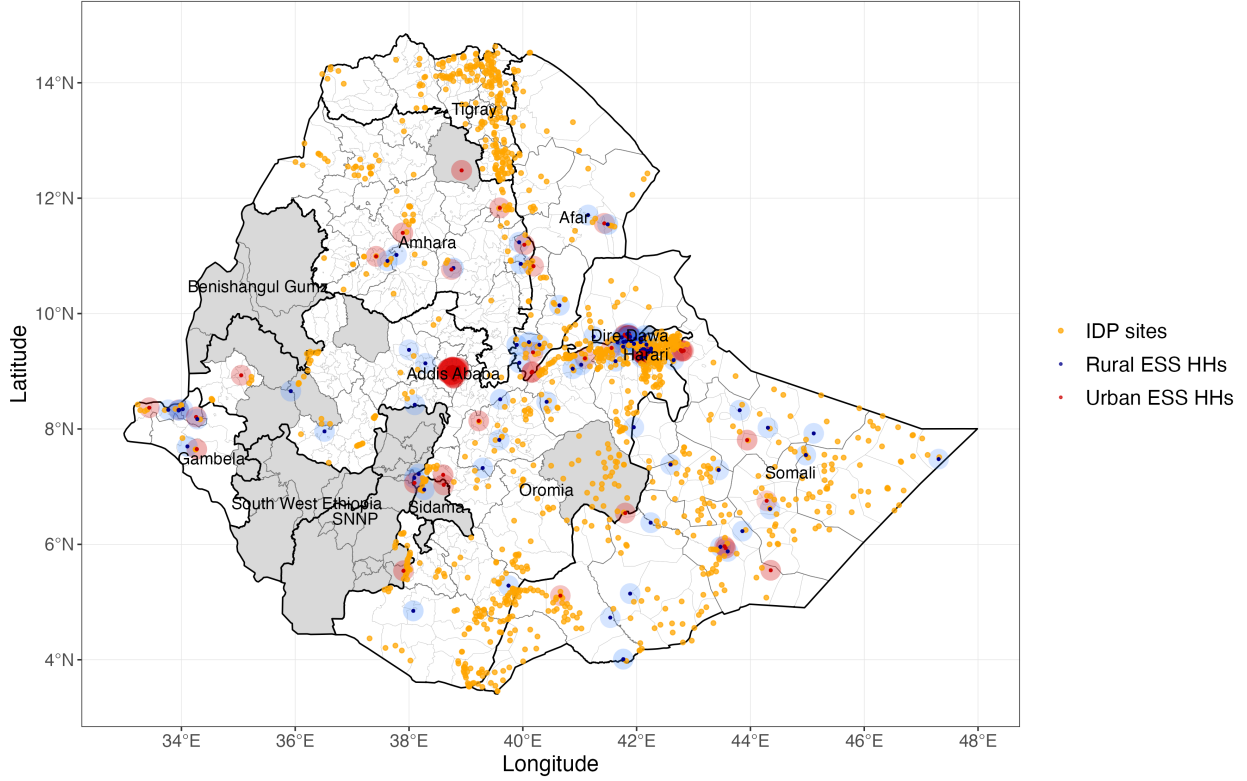
<i>Panel A: Rural households, ESS household survey</i>		
	Households	Individuals (15+)
Unrestricted data	3,180	10,729
In DTM geographic coverage	2,528	8,571
Exposed to IDPs in fixed-share baseline	1,290	4,431
Observed twice (HH FE support)	795	3,115
<i>Panel B: Urban households, ESS household survey</i>		
	Households	Individuals (15+)
Unrestricted data	3,700	10,779
In DTM geographic coverage	2,894	8,492
Exposed to IDPs in fixed-share baseline	2,035	5,974
Observed twice (HH FE support)	1,389	4,589
<i>Panel C: Rural households, ESS post-planting survey</i>		
	Households	Individuals (15+)
Unrestricted data	3,110	9,839
In DTM geographic coverage	2,342	7,742
Exposed to IDPs in fixed-share baseline	1,107	3,701
Observed twice (HH FE support)	534	2,012

Notes: Cells report unique counts, not household-year observations. The Households column counts distinct households; the Individuals column counts distinct ESS respondents age 15 or older in those households. Rows are nested sample restrictions. Unrestricted data includes households observed at least once in the listed survey-household-type sample. In DTM geographic coverage restricts to households in region-zone pairs covered by conflict-displacement DTM site assessments with GPS in 2019 or 2022, matching the geographic coverage restriction in the merged analysis files. Exposed to IDPs further restricts to households with positive nearby IDP support within 20 km in the fixed-share baseline: R11 (June 2018) for the household survey panels and R9 (January 2018) for the post-planting panel. Observed twice further restricts to households observed in both Wave 4 and Wave 5 within the DTM-coverage and baseline-exposed sample, corresponding to household fixed-effects support. For the post-planting survey, individuals are rostered respondents age 15 or older in households appearing in the post-planting sample.

The food security measures are reported in both of the forms described above, standardized and expressed in within-wave percentiles, and for both the full and the common-item index. Reassuringly, the full and common-item versions are close to one another, in their means within each setting and in the rural–urban gap between those means, which suggests that restricting the index to the five dimensions common to the two waves discards little information. Both indices place rural households below urban ones on either scale, indicating lower food security in rural areas, and the two scalings agree on the ranking.

Labor markets differ at least as sharply. Among individuals aged 15 or older, 49% in rural areas had taken part in an income-generating activity in the preceding seven days, against 40% in urban areas. School enrollment does little to explain the gap, differing by only three percentage points across the two; agricultural work for one’s own household does far more, engaging 44% of rural

Figure 3: Location of ESS households and IDP sites in June to August 2019



Note: Blue dots mark rural ESS wave 4 household-survey locations and red dots mark urban ESS wave 4 household-survey locations that enter the final household-survey analysis sample. The sample keeps households observed in both ESS waves 4 and 5 with positive nearby IDP support within 20 km in the R11 fixed-share baseline. Faint circles show 20 km buffers around those ESS households. Orange dots mark IDP sites from DTM site assessment round 17. Greyed-out areas are zones without DTM site-assessment coverage in the 2019 map layer; white areas are zones with DTM coverage, with woreda boundaries shown in light grey and regional boundaries shown in black.

individuals against 5% of urban ones. The composition of the remaining activity runs the other way, with self-employment three times as common in urban areas (12% against 4%) and wage employment more than seven times as common (22% against 3%). Hours worked are measured unconditionally and track these participation patterns closely.

The two settings differ along demographic and educational lines as well. Rural households are larger and marginally younger, at 33 against 34 years on average, while the sample's gender ratio is skewed towards women in urban areas. Literacy stands at about 44% in rural areas against 82% in urban areas, and primary school completion at roughly 14% against 60%, even though the share of individuals who ever attended school is very similar in the two. Conflict, finally, is more prevalent in urban areas, which may reflect their population density, about seven times the rural figure in this sample, as Table 2 shows.

Table 4 turns to the agricultural analysis sample, restricted to households active in crop cultivation.

Table 2: Summary Statistics for one-year flow exposure of hosts households to IDPs

Variable	N	Mean	SD	Min	Max
<i>Panel A: Rural households</i>					
One-year IDP flow exposure (1,000s)	1024	3.47	8.71	-12.58	65.85
Population within 20 km of EA centroid (1,000s)	1024	357.20	304.63	8.50	890.21
One-year IDP flow / 20 km population (%)	1024	2.71	12.85	-22.76	133.65
<i>Panel B: Urban households</i>					
One-year IDP flow exposure (1,000s)	1082	5.39	7.70	-3.48	56.70
Population within 20 km of EA centroid (1,000s)	1082	475.77	335.35	13.01	3057.59
One-year IDP flow / 20 km population (%)	1082	2.29	4.47	-6.43	27.45
<i>Panel C: Rural households, post-planting exposure</i>					
Eight-month IDP flow exposure (1,000s)	1035	2.81	6.50	-10.49	33.86
Population within 20 km of EA centroid (1,000s)	1035	440.17	312.63	11.93	880.03
Eight-month IDP flow / 20 km population (%)	1035	0.52	1.73	-9.74	9.68

Notes: Statistics are computed at the ESS household-year level, using one observation per household-year from Ethiopia ESS waves 4 and 5, and are restricted to the final analysis sample. The final analysis sample keeps households observed in both Wave 4 and Wave 5 with positive nearby IDP support within 20 km in the fixed-share baseline: R11 (June 2018) for the household survey panels and R9 (January 2018) for the post-planting panel. Panels A and B use the 20 km GPS individual analysis dataset and report the one-year flow of IDPs within 20 km of the household, measured in thousands of IDPs and including IDP sites from host communities and camps. Panel C is restricted to rural observations in the 20 km GPS agricultural analysis dataset and reports the eight-month post-planting flow exposure used for post-planting agricultural outcomes. Missing IDP-flow values are coded as zero before scaling, following the treatment construction used in the analysis scripts. Relative treatment is computed using data from WorldPop, which estimates population density per grid cell at a resolution of 30 arc seconds, approximately 1 km at the equator, adjusted to match the corresponding official United Nations population estimates. The data used are for Ethiopia in 2017. Population within 20 km of the EA centroid is reported in thousands. In each panel, the percentage row divides the corresponding IDP-flow measure by the WorldPop 2017 population within 20 km of the EA centroid and multiplies by 100.

Over the season, these households devote an average of 301 hours of family labor to land preparation and the tasks associated with it, and hire 6.9 person-days for the same work; among the minority of household-years that hire at all, the daily wage paid averages 8.7 USD. Cultivation of staples is close to universal at 93 percent, while 59 percent of households grow cash crops.³ Asked what renting out a hectare of their agricultural land for the year would be worth, households put the figure at 2,252 USD on average.

³Staple crops as found in the ESS data: Barley, Maize, Millet, Oats, Rice, Sorghum, Teff, Wheat, Mung bean / masha, Cassava, Chickpeas, Haricot beans, Horse beans, Lentils, Field peas, Vetch, Gibto, Soya beans, Red kidney beans, Apples, Bananas, Lemons, Mandarins, Mangos, Oranges, Papaya, Pineapples, Citron, Beetroot, Cabbage, Carrot, Cauliflower, Garlic, Kale, Lettuce, Onion, Green pepper, Potatoes, Pumpkins, Sweet potato, Tomatoes, Godere / taro, Guava, Peach, Spinach, Enset, Watermelon, Avocados, Other root crop, Amboshika, Kazmir, Strawberry, Shiferaw / moringa, Other fruits, Other pulses, Other cereal, Other vegetable. Cash crops: Fennel, Linseed, Ground nuts, Nueg / nigerseed, Rapeseed, Sesame, Sunflower, Black cumin, Black pepper, Cardamom, Chilies, Fenugreek, Ginger, Red pepper, Turmeric, White cumin, Chat, Coffee, Cotton, Gesho, Sugar cane, Tobacco, Coriander, Sacred basil, Rue, Gishita, Timiz kimem, Other spices, Other oilseed.

5 Empirical Strategy

5.1 Estimating equation

The object of interest is the effect of recent forced displacement on host households, estimated separately for rural and urban hosts. Recovering it requires first settling on a definition of displacement and on the dimension of it that plausibly bears on hosts. I measure displacement as a flow over a fixed window rather than as a stock, which places recent arrivals at the center of the analysis and keeps the estimates from being dominated by long-standing displaced populations. This choice also does useful work for identification, since it frees past displacement patterns to serve as an instrument rather than as the treatment itself, which is what the enclave design described below requires.

Concretely, and as set out in Equation 32, the treatment in the household analysis sample is the net number of IDPs located within 20 km of a host household over the year preceding its observation. Data limitations discussed above prevent the same window from being used on the agricultural side, where Equation 33 shortens it to eight months.

Because this exposure is built from a circle drawn around the centroid of an enumeration area, it varies at the level of the EA and the survey round rather than at the level of the individual household: every sampled household in a given EA faces the same measured inflow. The notation below reflects this, indexing the treatment by the enumeration area a in which a household resides.

The unit at which outcomes are observed is not the same across the outcomes of interest, and the specification follows the data rather than imposing a common unit. Expenditures, food prices, and every agricultural outcome are recorded for the household as a whole, and are estimated as

$$y_{ht} = \beta_R \Delta IDP_{at} \times R_a + \beta_U \Delta IDP_{at} \times U_a + \mathbf{X}'_{ht} \Pi + \lambda_h + \delta_t + \varepsilon_{ht}, \quad (34)$$

where h indexes host households, a the enumeration area in which household h resides, and t time. Labor supply, by contrast, is recorded for each household member: participation in any economic activity, in agricultural work for the household, in self-employment and in permanent wage employment, together with the hours devoted to each and with total hours in any income-generating activity, are all attributes of a person rather than of a household. These outcomes are therefore estimated on individuals aged 15 or older, indexed by i within household h :

$$y_{iht} = \beta_R \Delta IDP_{at} \times R_a + \beta_U \Delta IDP_{at} \times U_a + \mathbf{W}'_{iht} \Gamma + \lambda_h + \delta_t + \varepsilon_{iht}. \quad (35)$$

In both equations the indicator U_a equals one for urban enumeration areas and zero otherwise, and $R_a = 1 - U_a$ indicates rural ones; the urban–rural classification, like exposure itself, is an attribute of the EA. Interacting the treatment with each indicator in turn, rather than including it once alongside a single interaction, lets both coefficients be read directly as level effects: β_R is the effect of displacement on rural hosts and β_U the effect on urban hosts. Whether the two settings respond

differently is then a test of $\beta_R = \beta_U$ rather than a test on a single coefficient.

The two specifications carry different controls, and the vectors $\mathbf{X}_{ht}$ and $\mathbf{W}_{iht}$ are correspondingly distinct objects with their own coefficient vectors. In the household-level specification, $\mathbf{X}_{ht}$ contains the demographic composition of the household—its size, the mean age of its members, the share of males, mean literacy and mean primary school completion—alongside measures of local conflict. In the individual-level specification, $\mathbf{W}_{iht}$ replaces these averages with the individual’s own age, gender, literacy and primary school completion, retaining household size as a household-level covariate. Both specifications control for conflict within 20 km of the EA centroid, measured as the number of conflict events, the number of events with fatalities and the number of events involving civilian targeting. Aggregating individual characteristics is thus a consequence of the outcome being household-level, not a requirement of the estimator: as Section 5.4 makes clear, the shift-level formulation used for inference accommodates either unit of observation.

The remaining terms are household fixed effects λ_h , which absorb any time-invariant confounder at the household level or coarser—including everything fixed about the enumeration area, since sampled households do not move between waves—and time fixed effects δ_t , which absorb trends common to all observations. Because the treatment is constant within an EA and the panel is balanced across the two waves, demeaning within household leaves exactly the same variation as demeaning within EA: identification of β_R and β_U rests on the change in an EA’s exposure between the two rounds. Household fixed effects are nonetheless retained in the individual-level specification, where they absorb household-specific levels that the EA effects would not.

Estimating either equation by ordinary least squares would not recover the causal effect of exposure to displacement, however well specified the controls and fixed effects. The difficulty familiar from the migration and displacement literature is that IDPs do not settle at random across potential destinations: places that receive more of them may differ from places that receive fewer in ways that also move host outcomes, and may differ in ways that change over time and so survive the household fixed effects. Host communities adopting increasingly welcoming policies towards displaced populations, for instance, would attract larger inflows while their outcomes evolve for reasons of their own, leaving the OLS estimate contaminated by simultaneity and by omitted variation that no available control captures.

5.2 Instrument and rationale

To address this, the analysis turns to a shift-share instrumental variable design identified through the exogeneity of the shocks rather than of the shares (Borusyak et al., 2022). The instrument follows the enclave logic: it exploits the tendency of displaced people to move towards destinations where earlier arrivals from their own area of origin have already settled, and it combines, for each destination, the composition of its pre-existing IDP population by woreda of origin with the conflict shocks subsequently experienced in those origin woredas.

For the household analysis dataset the instrument reads

$$Z_{at}^{HH} = \sum_o s_{ao} \times c_{ot}^{1yr}, \quad (36)$$

where s_{ao} is the share of the baseline IDP population around enumeration area a that originates from woreda o , and c_{ot}^{1yr} is the number of conflict events recorded in woreda o over the year preceding the observation of a at t . The shares are held fixed at a baseline predating the exposure window, DTM round 11 in June 2018, one year before the first ESS observations used here; Figure 2 situates that baseline relative to the survey and assessment rounds. Like the treatment, these shares are built from the DTM sites lying within 20 km of the EA centroid, and are therefore common to every household in an EA and to every member of those households. The agricultural analysis dataset uses the same construction with the shorter window applied to the shocks,

$$Z_{at}^{Ag} = \sum_o s_{ao} \times c_{ot}^{8mths}, \quad (37)$$

where conflict events are summed over eight rather than twelve months and the shares are fixed in January 2018. In both cases the shocks are recentered before entering the regression, following [Borusyak et al. \(2022\)](#), so that identification rests on deviations of origin-woreda conflict from its expected level rather than on its raw count.

The instrument thus predicts that a destination receives more IDPs when the origin woredas of its established displaced population have recently become more violent. Put differently, it uses historical settlement patterns to determine which distant conflicts a given host community is exposed to, and the timing of those conflicts to determine when. What identification requires is not that hosts with large or particular baseline enclaves be comparable to others, but that the conflict shocks arriving through those enclaves be as good as randomly assigned.

Figure 4 displays the variation underlying the 2022 instrument, with conflict intensity shaded in blue and the origin woredas of IDPs outlined in red. Outside the Somali and Oromia regions in the southeast, where out-migration is common but measured conflict is low, conflict and displacement are widespread across the country, so the instrument does not draw on a single region. Table 5 reports the corresponding summary statistics at the origin-woreda level. The average origin woreda records 0.83 conflict events in the year before the IDPs from it are observed at destination, and supplies displaced people to 37 host households on average, with a mean baseline stock of 2,441 IDPs. The mean exposure share is 0.62, indicating that the displaced population surrounding a typical host EA is drawn from a small number of origins; the lower this share, the more distinct origin shocks contribute to an area's instrument.

This concentration is in part an artefact of the data rather than of displacement patterns themselves. The DTM records only the woreda of origin of the majority of the IDPs at a given site, so a site hosting people displaced from several woredas enters the analysis as though its entire population

came from one. The set of origins available for constructing the shares is correspondingly narrower than the true set, and the shares themselves are more concentrated than the underlying enclaves. Two consequences follow. The number of distinct shocks entering the instrument is smaller than the number of woredas actually sending IDPs to these destinations, which matters for inference in the shift-level formulation of Section 5.4, where the asymptotics run in the number of shocks; Section 6 accordingly reports the effective number of shocks alongside the first-stage diagnostics. The measured shares are also subject to error, in that minority origins are assigned a share of zero and the majority origin absorbs their weight. Because identification here rests on the exogeneity of the shocks rather than on the shares being correctly measured (Borusyak et al., 2022), this second point need not threaten consistency, though it does mean the instrument uses less of the available variation than a complete origin composition would allow.

5.3 Threats to identification

Instrumenting in this way confines the estimation to the component of displacement predicted by conflict in the origin woredas of the baseline enclave, interacted with the composition of that enclave. Variation arising at destination—local conditions that make a community more or less attractive to arriving IDPs—is discarded, which addresses the simultaneity concern raised above. What remains at issue is the exclusion restriction, namely that origin-woreda conflict reaches host outcomes only by way of the displacement it induces. The fixed effects narrow this considerably: with household and time effects absorbing level differences and common trends, a violation must take the form of a differential trend linking origin conflict to host outcomes.

The most immediate candidate is conflict itself travelling through space, particularly where the armed groups active at origin also operate near the destination. Conflict at destination is therefore included among the controls, in the form of the number of conflict events, the number of events with fatalities, and the number of events with civilian targeting within 20 km of the host EA centroid, so that any contemporaneous violence around the host community is held constant.

A second candidate is trade. If conflict at origin disrupts the routes connecting origin and destination woredas, food prices at destination would move for reasons unrelated to the arrival of IDPs, and increasingly so as origin conflict intensifies. The sign of that movement is determined by the direction in which food travels along the disrupted route. Where origin woredas supply food to a destination, disruption restricts the quantity reaching it and pushes prices there up; where origin woredas instead buy food from a destination, disruption strands that output at home and pushes prices there down. Pairing each case with what the host area itself produces gives the channel a testable signature. Rural host areas produce food and would be the exporters in such a relationship, so trade disruption predicts falling prices there; urban host areas produce very little of what they consume and would be the importers, so it predicts rising prices. The estimates reported below run in the opposite direction in both cases: food prices rise in rural host areas, indicating excess demand rather than the excess supply that disrupted exports would create, and fall in urban ones.

This pattern is difficult to reconcile with trade disruption and points instead to a channel operating through the displaced population itself, though it remains evidence read off the results rather than an independent test.

5.4 Clustering and shift-level estimation

A defining feature of this design is that a single shock in an origin woreda propagates to every destination hosting IDPs from it. Host households are therefore not independent draws: those sharing an origin in their baseline enclave share exposure to the same shock, and inference that ignores this dependence will overstate precision. The natural response, clustering at the origin-woreda level, is awkward in the specifications above, because the identifying variation lives at the origin-year level while the observations do not. Following [Borusyak et al. \(2022\)](#) (hereafter, BHJ 2022), I therefore estimate the equivalent regression at the level of the shocks. For the rural–urban specification, this means splitting the endogenous treatment into its rural and urban components and applying the same split to the instruments.

The procedure is set out below for the individual-level specification of Equation 35; the household-level case of Equation 34 is identical with the index i suppressed and $\mathbf{X}'_{ht}\Pi$ in place of $\mathbf{W}'_{iht}\Gamma$, since there is then a single observation per household-year. It begins by partialling out the controls and fixed effects. Each of the outcome and the two endogenous treatment components, $v_{iht} \in \{y_{iht}, \Delta IDP_{at} \times R_a, \Delta IDP_{at} \times U_a\}$, is residualized as

$$v_{iht}^\perp = v_{iht} - (\mathbf{W}'_{iht}\Gamma + \lambda_h + \delta_t). \quad (38)$$

For a candidate parameter vector $\beta = (\beta_R, \beta_U)'$, the residual is

$$e_{iht}(\beta) = y_{iht}^\perp - \beta_R (\Delta IDP_{at} \times R_a)^\perp - \beta_U (\Delta IDP_{at} \times U_a)^\perp, \quad (39)$$

and these residuals are aggregated to the origin-year level using the exposure shares, separately for rural and urban hosts. Since the shares are an attribute of the enumeration area, the aggregation factors into a sum over areas of an area-level share multiplying the residuals of everyone living there:

$$\tilde{e}_{ot}^R(\beta) = \sum_a s_{ao} R_a \sum_{h \in a} \sum_{i \in h} e_{iht}(\beta), \quad (40)$$

$$\tilde{e}_{ot}^U(\beta) = \sum_a s_{ao} U_a \sum_{h \in a} \sum_{i \in h} e_{iht}(\beta). \quad (41)$$

An enumeration area therefore contributes to these origin-year moments in proportion to the number of sampled individuals it contains. The estimator then chooses β_R and β_U to make the origin-

year shocks orthogonal to the aggregated moments,

$$\hat{\beta} = \arg \min_{\beta} \bar{m}(\beta)' W \bar{m}(\beta), \quad \bar{m}(\beta) = \begin{pmatrix} \sum_{o,t} c_{ot}^{1yr} \tilde{e}_{ot}^R(\beta) \\ \sum_{o,t} c_{ot}^{1yr} \tilde{e}_{ot}^U(\beta) \end{pmatrix}, \quad (42)$$

with W a weighting matrix. This shift-level GMM formulation is numerically equivalent to the interacted shift-share IV specification estimated at the level of the observation, but because its moments are indexed by origin and year, it permits standard errors clustered at the woreda of origin. The corresponding estimation on the agricultural analysis sample is simpler, since it involves no rural–urban split and its outcomes are household-level throughout, and is described in [Appendix A.4](#).

5.5 Validation

The instrument is strongly relevant. [Table 6](#) reports the shift-level first stages for the two endogenous treatment components. Among rural hosts, a recentered shock one unit above its expected value—that is, one additional share-weighted conflict event in the origin woredas of the baseline enclave, relative to what those shares alone would predict—is associated with roughly 409 additional net IDPs within 20 km of a host enumeration area. Among urban hosts the corresponding figure is 298 net IDPs. Both coefficients are significant at the 1 percent level, and the difference in magnitude is consistent with the larger absolute inflows recorded around rural hosts in [Table 2](#).

The conventional diagnostics point the same way. Because the specification contains two endogenous regressors, relevance is not settled by a single first-stage statistic: the instruments could predict each treatment component while failing to separate the two. The Sanderson–Windmeijer F-statistics address this directly, testing whether each component remains identified once the other is partialled out, and they take values of 217.99 for the rural component and 41.34 for the urban one. The Kleibergen–Paap rk Wald statistic evaluates the system as a whole, testing the rank of the first-stage coefficient matrix under clustering at the woreda of origin; its value of 60.01, together with the accompanying rank test, indicates that the two instruments jointly identify the two treatment components rather than collapsing onto a single dimension of variation.

These statistics should nonetheless be read with some caution. Their reference distributions are derived under homoskedasticity, so comparing cluster-robust statistics to the usual critical values is a convention rather than a valid test ([Andrews et al., 2019](#)), and the thresholds at which the conventional t -test attains correct size are substantially higher than the familiar value of ten ([Lee et al., 2022](#)). More importantly in this setting, they are large-sample statistics computed on a design whose effective sample is small. The effective number of shocks, reported in the same table as the inverse Herfindahl of the aggregate exposure weights, is about 12 origin-year cells for the rural component and about 9 for the urban one, against nominal counts of 104 and 143. The identifying variation is therefore concentrated in a limited set of origin-year cells, and the estimates depend on the conflict dynamics of those cells to a degree that the nominal cell and cluster counts

conceal. As set out in Section 5, part of this concentration is attributable to the DTM recording only the majority woreda of origin at each site.

Because the asymptotic approximations underlying the statistics above run in the number of shocks, I supplement them with randomization inference following [Borusyak and Hull \(2023\)](#) (hereafter, BH 2023), which draws counterfactual shocks from the distribution used to recenter the instrument and builds the null distribution of the first-stage coefficients by repeated assignment. The resulting p-values, reported alongside the conventional ones in Table 6, remain significant at the 5 percent level for both components. Relevance therefore survives an inferential procedure that makes no appeal to a large number of shocks, which is the more demanding test given the effective sample sizes above. Randomization inference p-values are displayed on all results tables and generally align with clustered standard errors from [Borusyak et al. \(2022\)](#).

6 Results

Effects on consumption outcomes I begin with food prices, reported in Table 7 for the three price variables described in Section 5, estimated separately for rural and urban hosts. Displacement produces local inflation in rural areas and deflation in urban ones, the sign pattern that Proposition 1 in Section 3 predicts when rural food markets are less integrated with outside markets than urban ones. Scaling the marginal effects by the average net inflow of IDPs over the preceding year, food prices in rural host areas rise by three to five percent relative to the national average, as columns 1 and 3 show. Urban host areas move in the opposite direction, with food prices falling by roughly four percent, though these estimates are less precisely determined. Columns 4 and 5, which use the standardized price index, confirm this divergence at least directionally. Section 7 takes up the reasons for it, which appear to involve food aid reaching urban areas considerably faster than rural ones.

Table 8 turns to expenditures. Total expenditures rise with displacement in both settings, by about 4 percent in rural areas and 3 percent in urban areas when scaled to the average shock. The composition of that increase differs sharply, and the difference tracks the price results. Nominal food expenditures rise in rural areas while the urban estimate is statistically indistinguishable from zero; deflating annual food expenditures by the woreda-level food deflator reverses the pattern entirely, leaving no detectable effect in rural areas and an increase of about 16 percent in urban ones. Rural households, in other words, spend more on food largely because food costs more, while urban households obtain more food for what they spend. Non-food expenditures rise by about 12 percent in rural areas when scaled, against a null effect in urban areas, though the absence of non-food price data in the ESS makes it impossible to separate quantity from price movements here.

Table 7: Effects of Displacement on Food Prices in Rural vs Urban areas

Prices:	Price deflator				Standardized price index	
	Woreda level		EA level		EA level	
	Rural	Urban	Rural	Urban	Rural	Urban
IDPs (thousands)	0.0141*** (0.0037) [0.021]	-0.0081* (0.0045) [0.123]	0.0067 (0.0042) [0.039]	-0.0079 (0.0052) [0.165]	0.0645*** (0.0123) [0.009]	-0.0402** (0.0169) [0.091]
Outcome mean	0.9952	0.9388	0.9791	0.9140	0.1320	-0.1215
Average # of IDPs (1k)	3.733	5.352	3.733	5.396	3.689	5.312
Implied effect (p.p. / S.D.)	5.27	-4.32	2.50	-4.25	0.238	-0.214
Observations	952	1,040	952	1,030	950	1,032
Households	476	520	476	515	475	516
Shifts	102	84	102	80	102	84

Notes: Each panel reports subgroup-specific BHJ 2022 shift-share IV/GMM estimates using the 1-year flow of IDPs within 20 km of LSMS households, measured in thousands and including IDPs in host communities and camps. The treatment and instrument are split into rural and urban components; the table reports the Rural and Urban subgroup slopes. Household and year fixed effects are included. Parentheses report BHJ 2022 shift-level standard errors clustered at the woreda of origin level. Square brackets report BH 2023 randomization-inference p-values. Stars use BHJ 2022 p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Columns 1 and 2 report the local food price deflator constructed from household food expenditure weights applied to local woreda-year community-market price ratios relative to national item averages. Columns 3 and 4 report the local food price deflator constructed from market prices in the community survey and merged at the EA level. Columns 5 and 6 report the standardized market-price index, also constructed from community-survey market prices and measured in standard deviations. For price-deflator columns, the implied effect equals 100 times the coefficient times the average number of IDPs, measured in thousands, and is reported in percentage points. For standardized price-index columns, the implied effect equals the coefficient times the average number of IDPs, measured in thousands, and is reported in standard deviations.

The food security estimates in Table 9 follow the same pattern and are reported for both indices and both scalings. Displacement worsens food security among rural hosts and improves it among urban ones, on every measure and at conventional levels of significance under randomization inference. Scaled to average exposure, rural households fall by roughly 0.06 standard deviations, or about two percentile points in the within-wave distribution, while urban households gain about 0.12 standard deviations, or between two and three percentile points. The full and common-item indices give almost identical estimates, which indicates that restricting the index to the five dimensions asked in both waves discards little information. As set out in Section 4, the recall period attached to the underlying questions changes between waves, and does so differentially across rural and urban areas. Because the indices are standardized within wave across both settings, this differential shift moves rural households down the wave-5 distribution relative to urban ones, in the same direction as the estimated effects. Appendix Table ?? therefore reproduces the specification with wave fixed effects interacted with the rural–urban classification, which absorbs any shift common to a setting and wave and so removes the artefact entirely. The estimates retain their signs and orders of magnitude, indicating that the pattern is not an artefact of the change in recall. Some caution remains warranted, since a measurement change that varies within setting would survive this correction, but the concern is considerably narrower than the raw comparison of affirmative

rates across waves would suggest.

Table 8: Effects of Displacement on Expenditures in Rural vs Urban areas

Outcomes:	Total	Food		Non-food	Food security
		National prices	Local prices		
<i>Rural households</i>					
IDPs (thousands)	78.7*** (29.0) [0.025]	62.2** (28.4) [0.038]	19.4 (66.2) [0.611]	30.3*** (5.8) [0.013]	-0.0169*** (0.0046) [0.017]
Outcome mean	7,480.3	5,977.1	6,326.8	852.4	-0.0708
Average # of IDPs (1k)	3.474	3.474	3.733	3.474	3.474
Implied effect (%)	3.66	3.61	1.14	12.35	-
Observations	1,024	1,024	952	1,024	1,024
Households	512	512	476	512	512
Shifts	104	104	102	104	104
<i>Urban households</i>					
IDPs (thousands)	44.2*** (14.4) [0.025]	13.7 (34.5) [0.550]	222.8** (91.6) [0.125]	15.5 (13.0) [0.128]	0.0232*** (0.0062) [0.034]
Outcome mean	9,177.9	6,200.4	7,392.3	1,538.1	0.0670
Average # of IDPs (1k)	5.386	5.386	5.352	5.386	5.386
Implied effect (%)	2.60	1.19	16.14	5.44	-
Observations	1,082	1,082	1,040	1,082	1,082
Households	541	541	520	541	541
Shifts	84	84	84	84	84

Notes: Each panel reports subgroup-specific BHJ 2022 shift-share IV/GMM estimates using the 1-year flow of IDPs within 20 km of LSMS households, measured in thousands and including IDPs in host communities and camps. The treatment and instrument are split into rural and urban components; the table reports the Rural and Urban subgroup slopes. Household and year fixed effects are included. Parentheses report BHJ 2022 shift-level standard errors clustered at the woreda of origin level. Square brackets report BH 2023 randomization-inference p-values. Stars use BHJ 2022 p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Expenditure outcomes are annual household aggregates in constant 2021 international dollars. Columns 1, 2, and 4 are reported in USD 2021 PPP. Column 3 reports annual food expenditures adjusted by the household-expenditure-weighted woreda-year local food price deflator. It is not adjusted using the EA-level community-survey deflator.

Taken together, the consumption results indicate that hosts in neither setting are made worse off in nominal terms, but that urban hosts gain considerably more in real terms. The price results account for the divergence: rural hosts face food inflation that absorbs much of their nominal gain, while urban hosts face falling food prices that amplify theirs. The food security estimates point the same way, and although the measures underlying them are imperfectly comparable across waves, they are imperfect in ways that the price and expenditure results do not share. That three sets of outcomes, resting on different survey modules and exposed to different sources of measurement error, agree on the direction of the rural–urban contrast makes it harder to attribute the pattern to any one of them.

Table 9: Effects of Displacement on Food Security Indices

Outcomes:	Full index		Common-items index	
	Standard deviations	Percentile	Standard deviations	Percentile
<i>Rural households</i>				
IDPs (thousands)	-0.0169*** (0.0046) [0.017]	-0.624*** (0.144) [0.019]	-0.0155*** (0.0041) [0.010]	-0.581*** (0.116) [0.016]
Outcome mean	-0.0708	49.785	-0.0523	50.448
Average # of IDPs (1k)	3.474	3.474	3.474	3.474
Implied effect (%)	-	-	-	-
Observations	1,024	1,024	1,024	1,024
Households	512	512	512	512
Shifts	104	104	104	104
<i>Urban households</i>				
IDPs (thousands)	0.0232*** (0.0062) [0.034]	0.539*** (0.162) [0.036]	0.0219*** (0.0063) [0.036]	0.420*** (0.153) [0.033]
Outcome mean	0.0670	53.916	0.0495	53.221
Average # of IDPs (1k)	5.386	5.386	5.386	5.386
Implied effect (%)	-	-	-	-
Observations	1,082	1,082	1,082	1,082
Households	541	541	541	541
Shifts	84	84	84	84

Notes: Each panel reports subgroup-specific BHJ 2022 shift-share IV/GMM estimates using the 1-year flow of IDPs within 20 km of LSMS households, measured in thousands and including IDPs in host communities and camps. The treatment and instrument are split into rural and urban components; the table reports the Rural and Urban subgroup slopes. Household and year fixed effects are included. Standard errors are BHJ 2022 shift-level standard errors clustered at the woreda of origin level. Stars use BHJ 2022 p-values: *** p<0.01, ** p<0.05, * p<0.10. The HH price deflator uses household food expenditure weights applied to local woreda-year community-market price ratios relative to national item averages. The community-price deflator uses expenditure-weighted community-level item price ratios merged back to households at the EA level.

Effects on labor market outcomes Table 10 reports effects on hours of labor, in total and by sector. Rural individuals increase their labor supply substantially: scaled by the average treatment, hours in any income-generating activity rise by 13 percent from a mean of 12 hours per week. This increase is driven by household agricultural work, which rises by 26 percent when scaled. What the sectoral estimates make clear is that these additional hours are not drawn from other activities. Hours rise by 0.70 in household agriculture and by 0.07 in self-employment, and fall by an insignificant 0.05 in wage employment, so that the gain in agriculture is very nearly the gain in total hours: summing across sectors leaves 0.72 additional hours of work, against 0.70 in agriculture alone. Had rural individuals been reallocating time between activities, the sectoral effects would have offset one another and left total hours roughly unchanged. Instead the estimates imply that the pull into household agriculture was met by working more overall rather than by withdrawing labor from elsewhere. Scaled to the average shock, the sectoral effects correspond to a 25 percent increase in household agriculture, an 18 percent increase in self-employment, and a 10 percent

decline in wage employment. The latter two proportional effects appear large mainly because rural individuals devote very few hours to these sectors to begin with, averaging 1.2 and 1.6 hours per week respectively.

Table 10: Effects of Displacement on Hours of Labor by Type in Rural vs Urban areas

Outcomes: Units:	Any IGA	Ag work for HH	Self-employment	Wage employment
	Hours in the last 7 days			
<i>Rural households</i>				
IDPs (thousands)	0.499*** (0.090) [0.017]	0.696*** (0.074) [0.012]	0.070*** (0.021) [0.027]	-0.052 (0.064) [0.232]
Outcome mean	11.518	8.189	1.195	1.564
Average # of IDPs (1k)	3.024	3.024	3.024	3.024
Implied effect (%)	13.11	25.71	17.68	-9.96
Observations	1,997	1,997	1,997	1,997
Households	614	614	614	614
Individuals	1,997	1,997	1,997	1,997
Shifts	103	103	103	103
<i>Urban households</i>				
IDPs (thousands)	-0.028 (0.057) [0.430]	-0.157*** (0.021) [0.014]	0.026 (0.031) [0.326]	0.183* (0.098) [0.112]
Outcome mean	16.395	0.794	4.054	9.188
Average # of IDPs (1k)	3.993	3.993	3.993	3.993
Implied effect (%)	-0.69	-78.77	2.53	7.95
Observations	4,037	4,037	4,037	4,037
Households	1,197	1,197	1,197	1,197
Individuals	4,037	4,037	4,037	4,037
Shifts	85	85	85	85

Notes: Each panel reports subgroup-specific BHJ 2022 shift-share IV/GMM estimates using the 1-year flow of IDPs within 20 km of LSMS households, measured in thousands and including IDPs in host communities and camps. The treatment and instrument are split into rural and urban components; the table reports the Rural and Urban subgroup slopes. Household and year fixed effects are included. Parentheses report BHJ 2022 shift-level standard errors clustered at the woreda of origin level. Square brackets report BH 2023 randomization-inference p-values. Stars use BHJ 2022 p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. For this version, outcomes and the interacted treatment variables are residualized in the age-15-plus individual-level sample before joining fixed R11 enclave shares and aggregating the residualized variables to the shift level. Outcomes are individual hours worked in the last seven days.

Urban individuals respond quite differently. There is no evidence of a change in total hours worked; instead, labor shifts out of household agriculture and into wage employment. Hours in household agricultural work fall by 0.16 on average, a 79 percent decline when scaled by average exposure, while hours in wage employment rise by 0.18, an increase of about 8 percent. Both patterns are consistent with Proposition 2 in Section 3: a rise in the relative return to food production pulls rural hosts toward agriculture, while a fall in that return pushes urban hosts toward services.

Effects on agricultural production outcomes Table 11 turns to the agricultural analysis sample, reporting effects on whether a household is active in crop cultivation at all, and then,

among cultivating households, on hours of family labor, person-days of hired labor, and the daily wage paid to hired workers, all for land preparation and the associated tasks of planting, ridging, weeding and fertilizing.

Table 11: Effects of Displacement on Agricultural Labor Outcomes for Land Preparation

	Crop activity	Family labor	Hired labor	
	Active	Hours	Person-days	Daily wage
IDPs (thousands)	-0.004*** (0.001) [0.022]	57.213*** (0.643) [0.001]	0.179*** (0.028) [0.045]	0.113 (0.073) [0.364]
Outcome mean	0.933	301.245	6.922	8.709
Average # of IDPs (1k)	3.022	3.128	3.128	3.050
Implied effect (%)	-1.43	59.41	8.09	3.94
Observations	658	590	590	78
Households	329	295	295	39
Shifts	54	54	54	20
First-stage coefficient	0.089***	0.081***	0.081***	0.008***
First-stage SE	(0.021)	(0.017)	(0.017)	(0.003)
First-stage F-stat	17.81	21.97	21.97	9.77

Notes: The table reports main-specification BHJ 2022 shift-share IV/GMM estimates for rural agricultural labor outcomes in Ethiopia. The treatment is the 1-year flow of IDPs within 20 km of LSMS households, measured in thousands and including IDPs in host communities and camps. Parentheses report BHJ 2022 shift-level standard errors. Square brackets report BH 2023 randomization-inference p-values. Stars use BHJ 2022 p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Family labor at planting is total household family labor hours in the planting module, winsorized at the 95th percentile. Hired labor at planting is total hired labor person-days in the planting module, winsorized at the 95th percentile; the daily wage column reports person-day-weighted hired labor wages winsorized at the 5th and 95th percentiles and converted from ETB to USD 2021 PPP using annual CPI and the 2021 GDP PPP conversion factor. The active crop cultivation column reports the conditioning indicator, defined as reporting household crop activity, any cultivated plot, or a positive number of cultivated plots, on the full model-complete post-planting sample. Planting labor estimates are restricted to model-complete household-year observations with active crop cultivation. The Ethiopia LSMS agricultural modules provide agriculture outcome data for rural areas only, so this table is restricted to rural households with agriculture-module data. Implied effect (%) equals $100 \times \beta \times \text{treatment mean} / \text{sample mean of the outcome}$.

Column 1 shows that displacement slightly contracts the extensive margin: exposure reduces the probability of being active in crop cultivation by about 1.4 percent when scaled to the average inflow, from a base of 93 percent. The remaining columns are estimated on cultivating households and therefore describe the intensive margin. Restricting the sample in this way conditions on an outcome that the treatment itself affects, but the scope for selection is limited by the size of the effect in column 1: roughly one household in seventy is moved across the cultivation margin at mean exposure, and since the households induced to exit are those closest to reporting no agricultural labor at all, trimming them from the comparison group moves the mean of the hours distribution very little. Whatever selection remains would inflate rather than deflate the estimates that follow.

Among cultivating households, the results confirm the pull into agricultural work seen in the household survey and locate it on both margins of the farm labor input. Family labor rises steeply, by about 59 percent when scaled to the average exposure to IDPs, against a mean of 301 hours devoted to these tasks. Hired labor rises as well, though far more modestly, with person-days up about 8 percent from a mean of 6.9; no effect on the daily wage is detected, though that column rests on the 78 household-years that hire at all and on 20 shocks, and is better read as uninformative than as evidence of no effect. Taken together, the estimates describe a consolidation of agricultural activity: slightly fewer households cultivate, while those that continue expand the labor devoted to land preparation substantially, and do so principally by working more themselves rather than by buying labor in.

Displacement also affects what is produced, as Table 12 shows. While 93 percent of households in this sample cultivate staple crops against 59 percent for cash crops, farmers exposed to larger inflows shift away from cash crops and towards staples. The reported value of land rises steeply as well: asked what they would earn from renting out all of their agricultural land, farmers exposed to 1,000 additional IDPs expect an increase of USD 415 on average, against a sample mean of USD 2,252.

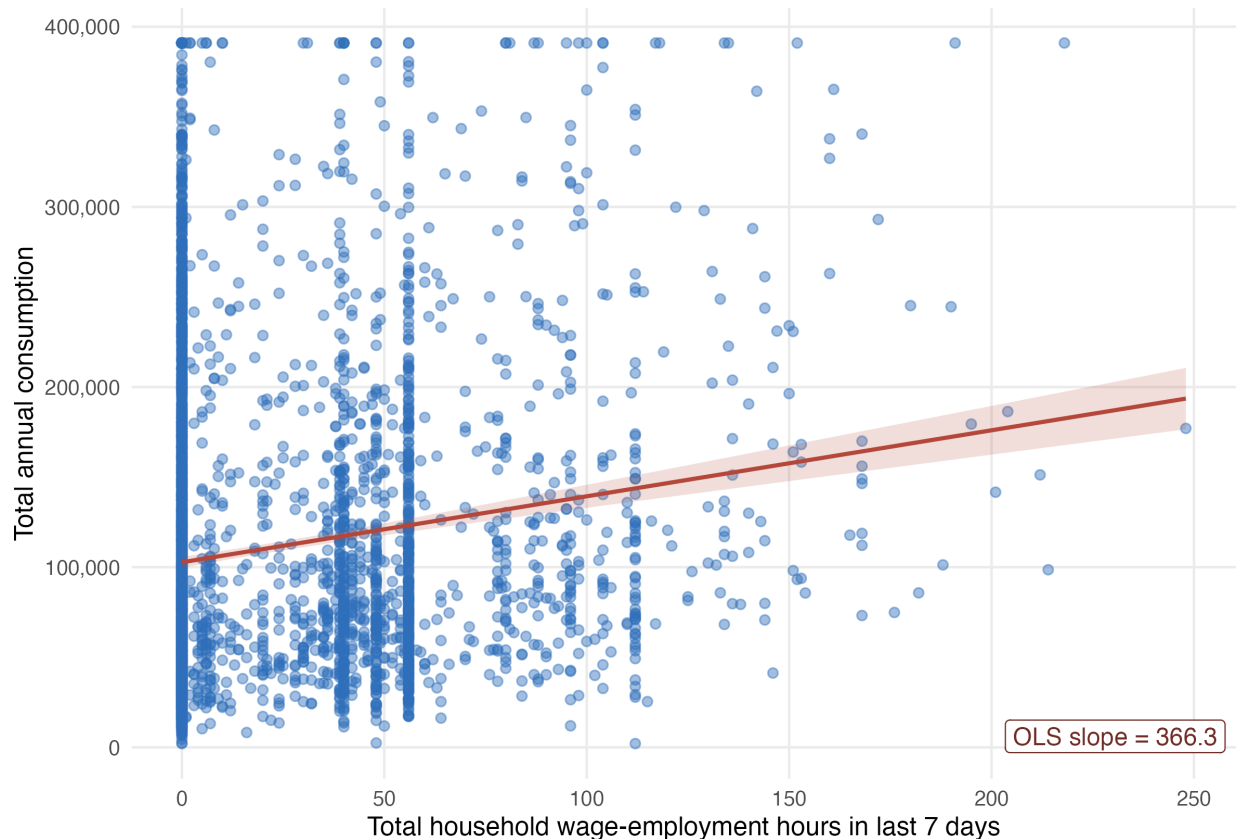
Table 12: Effects of Displacement on Agricultural Cultivation and Perceived Rental Rate of Land

	Cultivation		Land
	Staple crops	Cash crops	Rent/ha
IDPs (thousands)	0.018*** (0.002) [0.032]	-0.005*** (0.001) [0.018]	415.787*** (36.157) [0.052]
Outcome mean	0.927	0.588	2252.335
Average # of IDPs (1k)	2.696	2.696	2.838
Implied effect (%)	5.28	-2.39	52.39
Observations	582	582	374
Households	420	420	340
Shifts	46	46	46
First-stage coefficient	0.044***	0.044***	0.089***
First-stage SE	(0.009)	(0.009)	(0.020)
First-stage F-stat	22.27	22.27	19.30

Notes: The table reports main-specification BHJ 2022 shift-share IV/GMM estimates for rural cultivation and agricultural land outcomes in Ethiopia. The treatment is the 1-year flow of IDPs within 20 km of LSMS households, measured in thousands and including IDPs in host communities and camps. Stars use BHJ 2022 p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors are BHJ 2022 shift-level standard errors. Cultivation outcomes are indicators from the post-harvest crop module. Rent/ha uses the post-planting parcel question asking how much the parcel could be rented for. It is computed as the household mean of parcel-level potential rent divided by plot hectares among plots of at least 0.05 hectares, winsorized at the 5th and 95th percentiles and converted from ETB to USD 2021 PPP per hectare. The Ethiopia LSMS agricultural modules provide agriculture outcome data for rural areas only, so this table is restricted to rural households with agriculture-module data. Implied effect (%) equals $100 \times \text{beta} \times \text{treatment mean}$ divided by the sample mean of the outcome.

The rural results thus describe a strong pull into household agricultural work, and into staple crop production in particular, met largely by family labor and accompanied by higher expected land values. This constellation is consistent with a demand shock created by the arrival of IDPs, of which the food price increases documented above are the clearest expression. In urban areas, displacement instead moves hours out of agriculture and into wage employment, a pattern consistent with the occupational upgrading documented elsewhere in the literature and generally associated with higher incomes (Winton, 2025). Figure 5 plots total annual household expenditures against household hours of wage employment; the positive association suggests that the same relationship holds in urban Ethiopia, though it is a descriptive correlation rather than evidence that the reallocation raised incomes.

Figure 5: Scatterplot and regression line for total annual household consumption and total household hours in wage employment



Notes: The figure plots urban household-year observations from the ESS survey waves fielded in 2019 and 2022. The horizontal axis reports total household wage-employment hours worked in the seven days prior to the interview, constructed by summing individual wage-employment hours within each household-year after assigning missing individual hours to zero. The vertical axis reports total annual household consumption. The fitted line is from an ordinary least squares regression of total annual consumption on total household wage-employment hours, and the shaded band reports the corresponding confidence interval.

7 Mechanisms

The results above leave a puzzle. Displacement raises food prices in rural host areas and lowers them in urban ones, and the labor market responses diverge just as sharply, with rural hosts drawn into household agriculture while urban hosts move out of it and into wage employment. This section examines what distinguishes the two settings in ways that could produce such opposite responses to a common shock. Three candidates are considered in turn: differences in the reach of humanitarian food assistance, differences in market integration and access, and differences in the sectoral composition of employment, and in particular in levels of primary school completion - which proxies for skill and mobility across labor sectors. The three are not independent—an area poorly served by markets is also likely to be one where food aid doesn’t reach—but they carry distinct implications, and the evidence bears on each separately.

7.1 Food Aid and Market Access

The reversal in food price effects is the natural place to begin. That displacement raises food prices in rural areas indicates that the additional demand created by arriving IDPs is not met by a corresponding increase in local supply. Agricultural production responds slowly, since a season must pass before additional labor becomes additional output, so within the year over which the treatment is measured any offsetting increase in supply must arrive from outside the area—either through trade with other regions or through food assistance from humanitarian and governmental agencies. In the language of the model in Section 3, this is a low value of η_{FR} , the short-run supply elasticity of rural food, relative to η_{NR} for services—exactly the condition Proposition 1 requires for the demand shock to appear in prices rather than being absorbed by production. That urban host areas experience deflation over the same period suggests such a mechanism operates there. What follows asks whether it is present in urban areas and absent in rural ones.

Food aid. I exploit two features of the DTM data to test whether food assistance keeps pace with arrivals more readily in urban areas. The first is that, for every group of IDPs, key informants are asked to identify the main source of food available to them, choosing among food aid, donations from the host community, cash assistance in the form of transfers or vouchers redeemable at local markets, own cultivation or livestock, and other self-financed means of obtaining food. Following the spatial construction of the treatment described in Section 5, I compute the IDP-weighted average across all IDP groups located within 20 km of a host enumeration area for each of these sources, and estimate the shift-level specification with the resulting shares as outcomes.

Table 13 reports the estimates alongside the outcome means. Food aid is the main source of food for 70 percent of IDPs in rural areas and 78 percent in urban ones. In both settings, larger inflows reduce the share relying on it, which is what coverage failing to keep pace with arrivals would produce. The reduction is considerably larger in rural areas: an additional thousand IDPs over the year lowers the rural share by 4.3 percentage points against 1.5 points in urban areas, a coefficient nearly three times as large. Scaled to the average inflow in each setting, the rural share falls by about 16 percentage points, from 70 to 54 percent, against roughly 9 points in urban areas. The gap narrows under scaling, because average exposure is larger in urban areas, but the ordering is unchanged: assistance coverage erodes faster in rural areas as displaced populations grow.

Table 13: Effects of Displacement on Main Source of Food for IDPs within 20km of Host Households

Outcome:	Food aid	Host donation	Cash assistance	Cultivated livestock	Self-financed
<i>Panel A: Rural households</i>					
IDPs (thousands)	-0.0428***	0.0078	0.0100***	-0.0045**	0.0067***
BHJ 2022 SE	[0.0045]	[0.0054]	[0.0018]	[0.0020]	[0.0009]
Outcome mean	0.698	0.072	0.038	0.038	0.019
Average # of IDPs (1k)	3.586	3.586	3.586	3.586	3.586
Implied effect (p.p.)	-15.34	2.80	3.59	-1.60	2.39
Observations	654	654	654	654	654
Households	327	327	327	327	327
Shifts	92	92	92	92	92
Woredas of origin	62	62	62	62	62
<i>Panel B: Urban households</i>					
IDPs (thousands)	-0.0153***	-0.0017	0.0046**	-0.0079***	0.0022***
BHJ 2022 SE	[0.0023]	[0.0034]	[0.0018]	[0.0017]	[0.0008]
Outcome mean	0.783	0.028	0.054	0.032	0.022
Average # of IDPs (1k)	5.770	5.770	5.770	5.770	5.770
Implied effect (p.p.)	-8.81	-0.97	2.64	-4.56	1.26
Observations	898	898	898	898	898
Households	449	449	449	449	449
Shifts	123	123	123	123	123
Woredas of origin	62	62	62	62	62

Notes: The table reports total rural and urban treatment slopes from BHJ (2022) rural/urban interaction GMM specifications. Each outcome is the wave-specific IDP-population-weighted share of nearby IDPs within 20 km whose main source of food is the column category, using DTM site assessments in the LSMS interview year. The denominator is the nearby IDP population in host communities and camps with a known main food source. The treatment is the 1-year flow of IDPs in host communities and camps within 20 km, measured in thousands. The treatment and instrument are split into rural and urban components; the table reports the Rural and Urban subgroup slopes. Parentheses report BHJ (2022) shift-level standard errors clustered at origin woreda. Square brackets report BH 2023 randomization-inference p-values. Stars use BHJ (2022) p-values: *** p<0.01, ** p<0.05, * p<0.10. All specifications include household and year fixed effects plus the regular household control block: household size, demographics, education, and local conflict variables. Implied effects are percentage-point effects, equal to the coefficient times the average IDP flow in the corresponding rural/urban estimation sample, multiplied by 100.

Cash assistance moves in the opposite direction, reaching a larger share of IDPs in rural areas than in urban ones, again by roughly a factor of two. This is worth noting because cash and in-kind assistance have different price consequences. Where markets are thin and supply responds weakly, transfers that raise demand without adding local supply push prices up, an effect [Cunha et al. \(2019\)](#) document for cash transfers in remote Mexican villages. Rural host areas thus receive relatively more of the form of assistance that adds to inflationary pressure and relatively less of the form that offsets it.

These estimates speak to the extensive margin of assistance—the share of the displaced population reached—and not to its intensity. A falling share is consistent with total deliveries rising, if aid

volumes grow but more slowly than the population needing them, and it is the volume of food entering local markets that bears on prices. The second test therefore approaches the question from the price side.

Table 14: Effects of Displacement on Food prices for Aid-delivered vs Other Foods

Outcomes:	Humanitarian aid delivered foods		Crop foods	
	Aid-delivered	Other foods	Staple crops	Cash crops
Units:		Index		
<i>Rural households</i>				
IDPs (thousands)	-0.0092 (0.0062) [0.124]	0.0289*** (0.0035) [0.010]	0.0137** (0.0066) [0.026]	0.0306*** (0.0042) [0.013]
Outcome mean	0.947	1.011	1.067	0.987
Average # of IDPs (1k)	3.964	3.733	3.717	3.585
Implied effect (index)	-0.037	0.108	0.051	0.110
Observations	1,642	1,992	1,846	1,876
Shifts	90	102	100	102
<i>Urban households</i>				
IDPs (thousands)	-0.0162** (0.0065) [0.086]	-0.0066 (0.0041) [0.135]	-0.0123 (0.0076) [0.147]	-0.0144*** (0.0033) [0.027]
Outcome mean	0.929	0.934	0.962	0.911
Average # of IDPs (1k)	5.297	5.352	5.316	5.353
Implied effect (index)	-0.086	-0.035	-0.065	-0.077
Observations	1,642	1,992	1,846	1,876
Shifts	131	141	141	141

Notes: The table reports rural/urban heterogeneity in the effect of nearby IDP inflows on food prices. The treatment is the 1-year flow of IDPs in host communities and camps within 20 km, measured in thousands. The treatment and instrument are split into rural and urban components, and the table reports the Rural and Urban subgroup slopes. Columns report worda-level food price deflators for aid-delivered foods, other foods, staple crops, and cash crops. Implied effects equal the coefficient multiplied by the average IDP flow in the corresponding rural/urban estimation sample and are reported in index units. All specifications include household and year fixed effects and the regular control block used in the main consumption specification. Parentheses report BHJ (2022) shift-level standard errors clustered at worda of origin. Square brackets report BH (2023) randomization-inference p-values; stars use BHJ (2022) p-values: *** p<0.01, ** p<0.05, * p<0.10.

The DTM records which categories of food are distributed as assistance: cereals, pulses, oils, and corn soya blend. I group all ESS food items falling into these categories as aid-delivered foods and treat the remainder as other foods.⁴ I then reconstruct the local price deflator separately for each group and estimate the same specification on each, repeating the exercise for the staple and cash crop categorization used above.

Table 14 reports the results. Comparing aid-delivered foods with other foods confirms that assistance succeeds in restraining prices, in both rural and urban areas. The aid categories do not map

⁴Concretely: teff, wheat, maize, sorghum and rice for cereals; horsebeans, lentils and processed pulses (shiro) for pulses; and processed oils. The ESS records no corn soya blend.

cleanly onto the staple and cash crop distinction, however, and inflationary pressure remains for staple crops in rural areas. Column 2 is the most telling: for foods outside humanitarian assistance packages, urban areas contain the price pressure generated by displacement in a way that rural areas do not.

Read together, the two tables establish that food assistance matters for prices but does not arrive quickly enough in rural areas to absorb the demand created by new arrivals. They also point beyond assistance itself. Prices rise in rural areas for precisely those foods that assistance does not cover, which no account resting on aid alone can explain, and which suggests a broader difference in how the two settings obtain food from outside. Market integration is the natural candidate, and I turn to it next.

Market integration and access. Among the most salient differences between rural and urban settings in low-income countries is the degree to which local markets are integrated with markets elsewhere. The ESS community survey allows this to be measured directly: key informants report the distance from the community to a set of amenities that trade depends on, namely the nearest asphalt road, the nearest bus station, and the nearest large weekly market.

Table 15 reports these distances for rural and urban households in the analysis sample. The gaps are substantial. Asphalt roads are about 19 km closer for urban households than for rural ones, bus stations about 3 km closer, and large weekly markets about 7 km closer. To summarize the three, I take their unweighted average and refer to it as a market integration index; on this measure rural households are some 9 km further from the infrastructure of trade than urban ones.

Table 15: Measures of market access for rural and urban areas

Variable	Rural			Urban			Diff.
	N	Mean	SD	N	Mean	SD	
Nearest asphalt road (km)	1,024	28.16	56.15	1,082	9.48	23.34	18.68***
Nearest bus station (km)	990	10.11	27.01	1,077	6.83	35.62	3.28**
Nearest large weekly market (km)	1,024	7.86	11.21	1,082	1.28	4.28	6.58***
Market integration index (km)	1,024	15.30	24.92	1,082	5.85	13.13	9.45***
Harris MA index (1,000s)	990	73.59	77.53	1,077	177.03	206.56	-103.44***

Notes: Statistics are computed on household-years in the BHJ-supported Ethiopia total annual consumption household analysis sample. MA denotes market access. The MA average distance is the average distance to the nearest asphalt road, bus station, and large weekly market. Lower distances indicate better market access. The Harris MA index is computed within each survey year as the sum across other ESS enumeration areas of destination community population divided by the EA-to-EA centroid distance in kilometers, with theta set to 1 and the origin EA excluded. The table reports this measure only for household-years in the analysis sample and scales it in thousands. Difference is the rural mean minus the urban mean. P-values are from two-sided Welch tests of equality in means between rural and urban household-years. Significance stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I complement this index with a measure of market access in the tradition of [Harris \(1954\)](#), which

underlies more recent constructions in the trade and development literature ([Donaldson and Hornbeck, 2016](#)). For enumeration area a , the measure sums the population of every other enumeration area e , weighting each by the inverse of the distance between them, and is divided by one thousand for legibility:

$$MA_a = \frac{1}{1000} \sum_{e \neq a} \frac{Pop_e}{D(e, a)}. \quad (43)$$

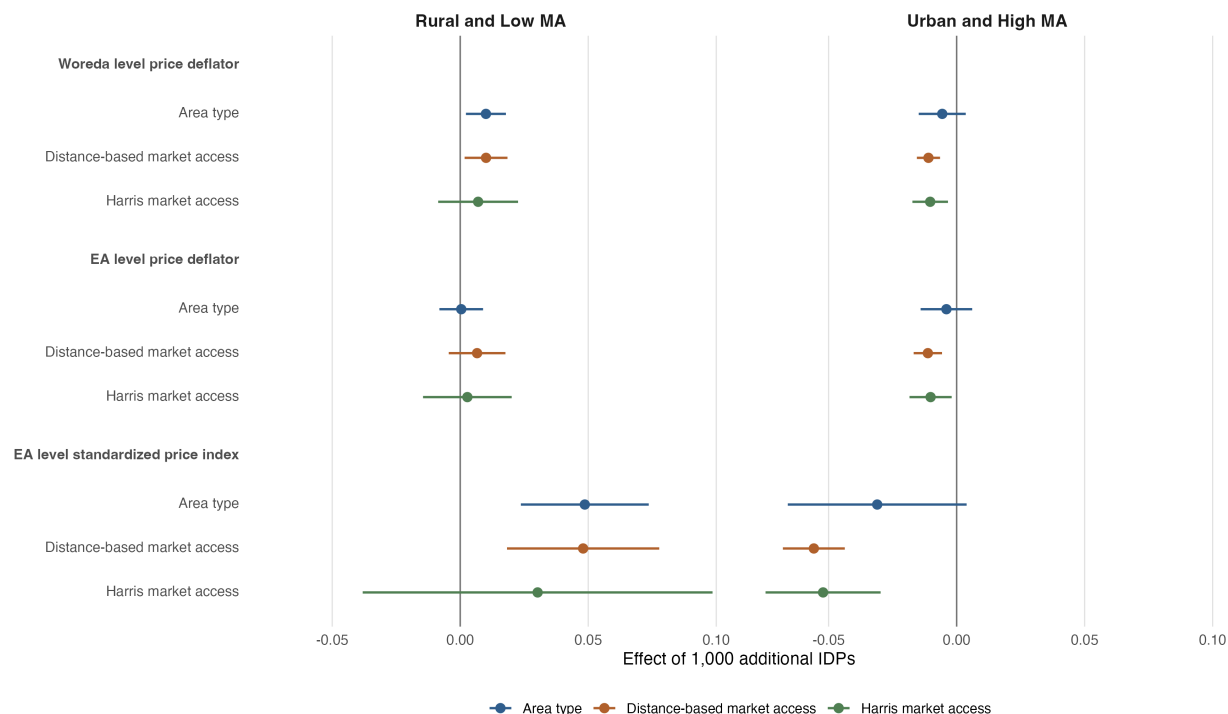
The two measures capture different things. The market integration index describes how far a community is from the physical infrastructure through which goods arrive, while the market access measure describes how much economic mass is within reach. The first is local and specific; the second is aggregate and spatial.

To test whether either accounts for the rural–urban contrast, I split the sample at the median of each measure and re-estimate Equations [34](#) and [35](#), replacing the rural–urban interaction with an interaction on the market variable. If market integration is what distinguishes the two settings, then splitting on it should reproduce the pattern that splitting on setting produces.

[Figure 6](#) reports the price results. Estimates for rural areas, for low market integration, and for low market access appear on the left, with their complements on the right, and estimates from the same specification are aligned on a common row. Both market splits reproduce the setting split closely: food prices rise with displacement in areas that are poorly integrated or poorly connected, and the effect is null to negative in areas that are well served, consistent with well-integrated markets behaving like the small open economy of [Section 3](#) rather than the autarky case. The market integration split is somewhat more precisely estimated than the market access split. [Figure 7](#) shows the same exercise for expenditures, where the correspondence is again close and, if anything, closer for market integration than for the rural–urban split itself.

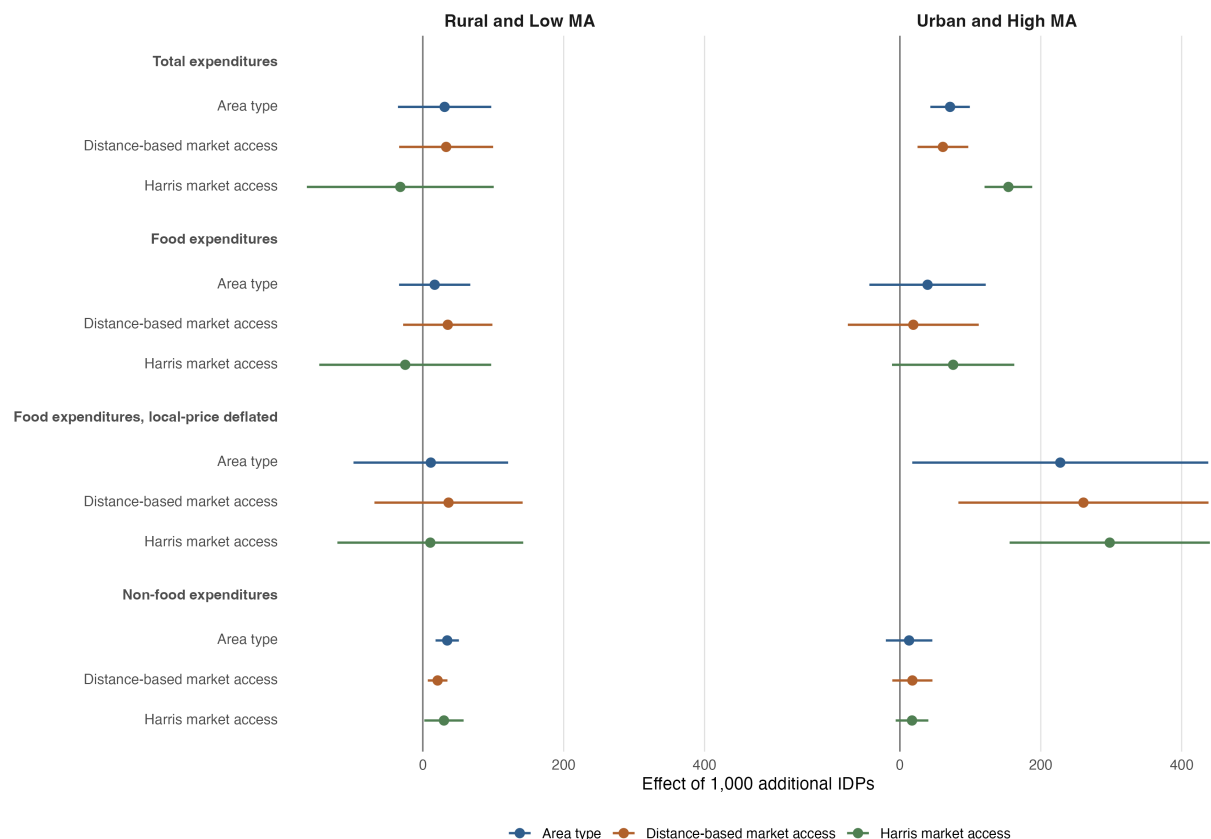
The labor market results in [Figure 8](#) are more mixed, and they separate the two measures. Splitting on market access does not reproduce the setting results at all, with signs flipping across several outcomes. Splitting on market integration does considerably better on the urban side of the contrast: it recovers the null effect on total hours, the shift away from household agricultural work, and the increase in wage employment, though the last is less precisely estimated. It also recovers the pull into household agriculture in poorly integrated areas. What it does not recover is the rest of the rural pattern, namely the increase in total hours and the null effect on wage employment.

Figure 6: The effect of displacement on food prices by area type, market integration and market access



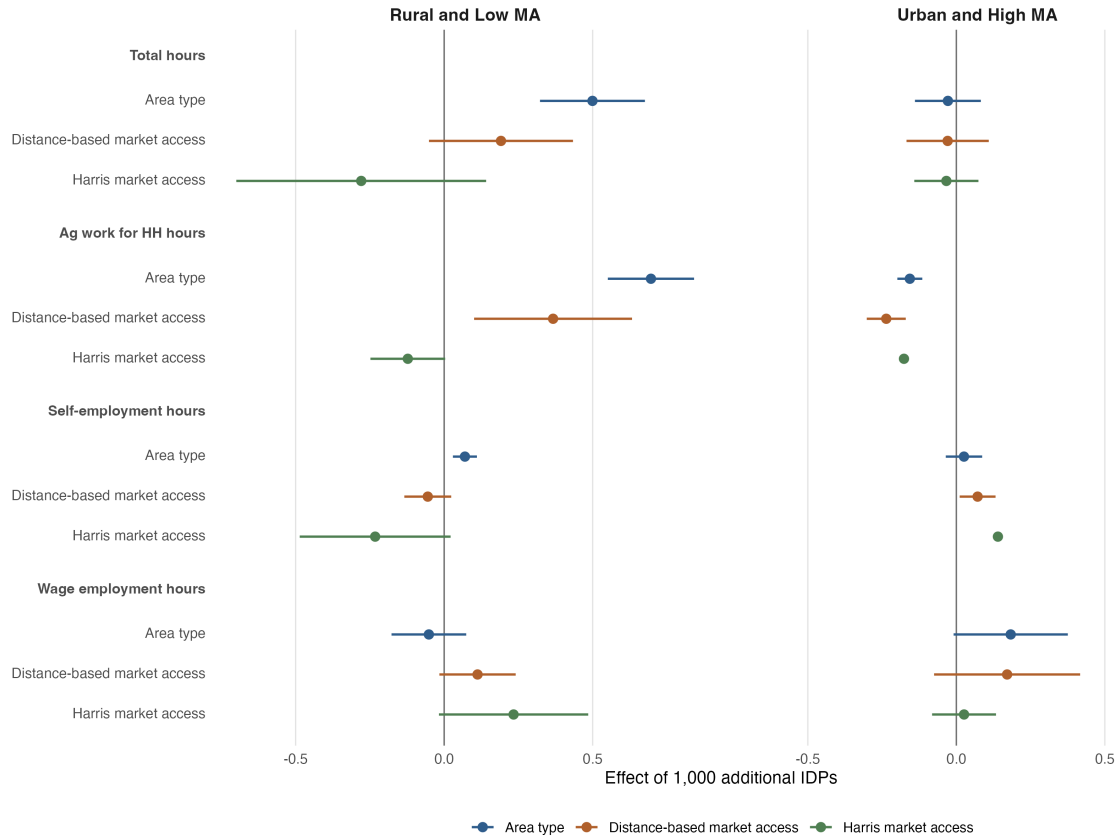
Notes: The figure reports BHJ (2022) estimates of the effect of 1,000 additional nearby IDPs on local price outcomes. The left panel plots estimates for rural households, low market integration, and low market access; the right panel plots estimates for urban households, high market integration, and high market access. The market integration index is defined using the average distance to asphalt roads, bus stations, and large weekly markets; high market integration corresponds to below/equal-median average distance. The market access index is the sum across all other ESS enumeration areas of destination population divided by centroid distance; high market access corresponds to above-median market access. Horizontal bars show 95 percent confidence intervals based on BHJ (2022) shift-level standard errors clustered at woreda of origin. Market-access estimates use household and year fixed effects and the regular aggregate-consumption control block.

Figure 7: The effect of displacement on expenditures by area type, market integration and market access



Notes: The figure reports BHJ (2022) estimates of the effect of 1,000 additional nearby IDPs on household expenditure outcomes. The left panel plots estimates for rural households, low market integration, and low market access; the right panel plots estimates for urban households, high market integration, and high market access. The market integration index is defined using the average distance to asphalt roads, bus stations, and large weekly markets; high market integration corresponds to below/equal-median average distance. The market access index is the sum across all other ESS enumeration areas of destination population divided by centroid distance; high market access corresponds to above-median market access. Horizontal bars show 95 percent confidence intervals based on BHJ (2022) shift-level standard errors clustered at woreda of origin. Market-access estimates use household and year fixed effects and the regular aggregate-consumption control block.

Figure 8: The effect of displacement on hours of labor by area type, market integration and market access



Notes: The figure reports BHJ (2022) estimates of the effect of 1,000 additional nearby IDPs on hours worked in the last 7 days. The left panel plots estimates for rural households, low market integration, and low market access; the right panel plots estimates for urban households, high market integration, and high market access. The market integration index is defined using the average distance to asphalt roads, bus stations, and large weekly markets; high market integration corresponds to below/equal-median average distance. The market access index is the sum across all other ESS enumeration areas of destination population divided by centroid distance; high market access corresponds to above-median market access. Horizontal bars show 95 percent confidence intervals based on BHJ (2022) shift-level standard errors clustered at woreda of origin. Market-access estimates residualize outcomes and interacted treatment variables at the individual-observation level using household and year fixed effects and the regular labor control block before aggregating residuals to shift-level moments.

Taken together, the two splits show that market integration accounts for a substantial part of what distinguishes rural from urban hosts, and does so without reference to the rural–urban classification at all. It explains the divergence in food prices and, through prices, the divergence in real expenditures. It also explains the reallocation of urban labor away from agriculture and the pull of rural labor into it. What it does not explain is why rural hosts work more in total rather than reallocating, nor why their wage employment is unaffected. Those features of the rural response appear to depend on something other than how well connected a community is to markets, and the following subsection takes up the composition of the workforce as a candidate.

7.2 Educational Attainment

Rural and urban labor markets differ in the schooling of the people employed. The contrast is stark: as Table 3 shows, 14 percent of rural individuals have completed primary school against 59 percent in urban areas. If schooling governs which sectors a worker can move into, the same shock will produce different aggregate responses in the two settings simply because the two workforces are composed differently. To test this, I estimate the effect of displacement on hours worked by sector across a two-by-two heterogeneity structure, interacting the rural–urban classification with an indicator for having completed primary school. Table 16 reports the results.

The rural estimates show that the pull into agriculture does not depend on schooling, as shown by the heterogeneous effects by market integration. Workers who have not completed primary school raise hours in household agricultural work by 0.32 per week, and workers who have raise them by 0.32 as well, corresponding to increases of 12 and 13 percent when scaled to average exposure. Total hours in income-generating activity rise by about 6 percent in both groups. Market integration seems to draws rural labor into agriculture reaches the educated and the uneducated alike, and in the same proportion.

Where the two groups differ is on the second margin. Rural workers who have completed primary school also raise hours in non-agricultural self-employment, by 0.15 or 22 percent when scaled, while the corresponding estimate for those who have not is small and indistinguishable from zero. This is the sense in which composition matters in rural areas. It is not that most rural workers respond differently to displacement, but that the additional margin available to educated workers is available to only 14 percent of them, so that in the aggregate the rural self-employment response is muted and rural labor appears to flow almost entirely into agriculture. Among individuals active in any income-generating activity, roughly 90 percent work in household agriculture in rural areas, against about 12 percent in urban areas (44 of 49 percent and 5 of 40 percent, respectively; Table 3). This is the low-density case discussed after Proposition 2 in Section 3: with few rural workers positioned near the occupational cutoff $\bar{z}_R$, the model’s selection margin is thin, and the rural response is concentrated on the intensive margin—more hours from workers already farming—rather than on reallocation across sectors.

The urban estimates describe the reverse movement, and here too it is common to both education groups. Workers without completed primary schooling reduce hours in household agricultural work by 0.26, and those with it reduce them by an almost identical 0.26; both raise hours in wage employment, by 0.18 and 0.31 respectively. Displacement in urban areas thus moves urban labor out of the small agricultural sector and into wage employment regardless of schooling, and in proportional terms the less educated gain more wage hours, 19 percent against 12, though from a much smaller base.

What separates the two urban groups is whether the reallocation is complete. Workers who have completed primary school end with total hours essentially unchanged, the hours lost in agriculture

being recovered in wage employment. Workers who have not end with total hours about 4 percent lower: the wage employment they gain does not fully replace the agricultural work they lose. The model in Section 3 anticipates the direction of this reallocation for both education groups but is silent on its completeness, since it does not sign the change in total hours for any one type; the asymmetry documented here is accordingly an empirical result rather than a model-implied one. The distributional consequence of displacement in urban areas therefore operates less through who moves into better-paid sectors—both groups do—than through who is able to move without leaving hours behind.

Table 16: The effects of displacement on hours of labor by area type and primary education completion

Outcomes: Units:	Any IGA	Ag work for HH	Self-employment Hours in the last 7 days	Wage employment
<i>Did not complete primary school - rural</i>				
IDPs (thousands)	0.199** (0.079) [0.014]	0.318*** (0.075) [0.000]	-0.016 (0.016) [0.319]	-0.031 (0.029) [0.289]
Outcome mean	10.620	9.195	0.682	0.352
Average # of IDPs (1k)	3.398	3.398	3.398	3.398
Implied effect (%)	6.36	11.76	-8.10	-29.73
<i>Completed primary school - rural</i>				
IDPs (thousands)	0.307*** (0.091) [0.001]	0.322*** (0.072) [0.000]	0.146*** (0.045) [0.002]	0.022 (0.094) [0.818]
Outcome mean	13.130	6.824	1.794	3.662
Average # of IDPs (1k)	2.767	2.767	2.767	2.767
Implied effect (%)	6.46	13.06	22.47	1.64
<i>Did not complete primary school - urban</i>				
IDPs (thousands)	-0.132*** (0.038) [0.001]	-0.260*** (0.043) [0.000]	0.015 (0.083) [0.857]	0.177** (0.079) [0.029]
Outcome mean	12.674	2.396	4.307	3.844
Average # of IDPs (1k)	4.034	4.034	4.034	4.034
Implied effect (%)	-4.20	-43.76	1.41	18.52
<i>Completed primary school - urban</i>				
IDPs (thousands)	-0.025 (0.053) [0.630]	-0.257*** (0.015) [0.000]	-0.011 (0.031) [0.730]	0.310*** (0.093) [0.001]
Outcome mean	17.170	0.427	3.702	10.744
Average # of IDPs (1k)	4.026	4.026	4.026	4.026
Implied effect (%)	-0.60	-242.58	-1.16	11.61

Notes: Each column reports a BHJ (2022) shift-share IV/GMM estimate for one hours-of-labor outcome. The treatment is the 1-year flow of IDPs within 20 km of LSMS households, measured in thousands and including IDPs in host communities and camps. The interaction splits adult individual-years into four cells by primary school completion and household rural/urban status. Outcomes and interacted treatments are residualized at the individual-year level with respect to household size, age, sex, literacy, school attendance, primary school completion, local conflict controls, household fixed effects, and year fixed effects before aggregation to shift-share cells. Standard errors are BHJ (2022) shift-level GMM standard errors clustered at origin woreda. Outcome means are computed within the corresponding subgroup estimation sample. Observations, individuals, and shifts by subgroup are: Did not complete primary school - rural: 3,721 observations, 2,441 individuals, 104 shifts; Completed primary school - rural: 630 observations, 484 individuals, 91 shifts; Did not complete primary school - urban: 2,225 observations, 1,689 individuals, 82 shifts; Completed primary school - urban: 2,859 observations, 2,310 individuals, 83 shifts. Stars use BHJ (2022) p-values: *** p<0.01, ** p<0.05, * p<0.10. The implied effect is the coefficient times the average treatment divided by the outcome mean.

Taken together, the four panels qualify the reading of the aggregate results offered in Section 6. The rural pull into agriculture and the urban shift towards wage employment are not phenomena confined to particular skill groups; they appear throughout the workforce in each setting. School-

ing determines the margins available alongside the dominant movement—an additional route into self-employment in rural areas, and full rather than partial replacement of lost hours in urban ones—and the very different educational composition of the two workforces then shapes what the aggregate response looks like. The occupational shift towards wage employment observed in urban areas is thus better described as broad-based, with the gains from it unevenly distributed, than as upgrading confined to educated workers. This pattern is consistent with a displacement literature that identifies winners and losers along skill lines, though it locates the difference in the completeness of the adjustment rather than in its direction.

7.3 Discussion

Of the three candidates examined in this section, market integration does the most work. Splitting the sample on how far a community sits from the infrastructure of trade reproduces the rural–urban contrast in food prices and in real expenditures without invoking the rural–urban classification at all, which suggests that what matters about being rural, for these outcomes, is largely being poorly connected. The chain runs from integration to prices and from prices to consumption: where additional demand cannot be met by goods arriving from elsewhere, it raises local food prices, and the resulting inflation absorbs the nominal expenditure gains that hosts would otherwise enjoy. Where goods do arrive, prices fall and the same nominal gains translate into real ones. This accounts for the increase in real consumption among urban hosts and for its absence among rural ones.

The same division carries onto labor markets, though less completely. Poorly integrated areas show the pull into household agriculture, and well integrated areas show the movement out of it and into non-agricultural self-employment and wage work. What the market integration split does not recover is why rural hosts raise their total hours rather than reallocating them, or why their wage employment is unaffected. These are features of the rural labor supply response that connectedness alone does not explain.

Food assistance operates on the same margin as market integration, and partly substitutes for it. Aid-delivered foods show restrained price effects in both settings, so assistance does attenuate the inflationary pressure that displacement creates. Its limitation is one of coverage rather than of effectiveness: the share of IDPs relying on food aid as their main source falls sharply as inflows grow, and it falls considerably faster in rural areas. Assistance therefore fails to hold rural food prices down not because it does not work where it reaches, but because it does not keep pace with arrivals where it is most needed. That prices rise in rural areas for precisely the foods outside assistance packages is consistent with this reading.

The composition of the workforce accounts for a further part of the picture, and specifically for who is able to move between sectors. The response to displacement within each setting is largely common across education groups, but the margins available alongside it are not. In rural areas, where only 14 percent of workers have completed primary school, the move into non-agricultural

self-employment is confined to those who have, so that in the aggregate rural labor appears to flow almost entirely into agriculture. In urban areas, where a majority have completed primary school, workers of all education levels shift hours from household agriculture into wage employment, but only those with schooling replace the hours they lose in full, while the less educated end up working less overall. Occupational upgrading is thus available in both settings and taken up in both, but its benefits are distributed according to schooling: in rural areas it is restricted to a small educated minority, and in urban areas it is broad-based but incomplete for those without primary education.

Read together, the three mechanisms are best understood as operating in sequence rather than in competition. Market integration determines whether the demand shock created by displacement is absorbed by supply from elsewhere or transmitted into local food prices. Food assistance can substitute for market integration, but only to the extent that its coverage keeps pace with arrivals, which in rural areas it does not. The composition of the workforce then determines how the resulting price and demand pressures are converted into labor market outcomes, and which hosts are positioned to benefit from them.

8 Conclusion

This paper has argued that the effects of displacement on host communities are best understood not as a single, generalizable outcome but as the joint product of two forces—a demand shock transmitted through consumer goods markets and a supply shock transmitted through labor markets—whose relative strength is shaped by local conditions. Applying a common empirical design to rural and urban destinations of internal displacement in Ethiopia reveals two starkly different host experiences arising from the same underlying shock. In rural areas, where market access is limited, displacement raises food prices, leaves real food expenditure unchanged, and draws family and hired labor further into agricultural production, particularly staple crops. In urban areas, where markets are better connected and humanitarian assistance covers a larger share of new arrivals, the same shock lowers food prices, raises real expenditure and food security, and shifts host labor out of household agriculture and into wage employment. Neither pattern reflects a fixed response to displacement as such; both follow from how well local food markets absorb an increase in demand, and from how the resulting price and labor market pressures interact with the skills already present in the host workforce.

These results speak to the two literatures the paper engages throughout. For the study of displacement’s effects on hosts, the fact that a single design, applied to a single country, reproduces both the negative consumption effects documented in poorly-connected rural settings ([Alix-Garcia and Saah, 2010](#); [George and Adelaja, 2022](#)) and the occupational upgrading documented in better-connected ones ([Winton, 2025](#)) suggests that much of the disagreement in this literature reflects differences in market integration, aid coverage, and workforce composition across settings, rather than differences in how displacement itself operates on hosts. For the literature on structural transformation, the results show that displacement can move workers out of household agriculture much as improve-

ments in market access or infrastructure have been shown to do ([Asher and Novosad, 2020](#); [Brooks and Donovan, 2020](#)), and that in urban Ethiopia this movement is already broad-based rather than confined to more educated hosts; what schooling governs there is only whether the hours lost in agriculture are fully replaced by wage work. In rural Ethiopia, by contrast, schooling determines whether a host gains an additional foothold in non-agricultural self-employment at all, and because just 14 percent of rural hosts have completed primary school, this diversifying margin is nearly invisible in the aggregate, leaving rural labor to appear, in the data, as flowing almost entirely into agriculture.

Beyond these contributions, the findings also carry a practical implication for humanitarian response: because rural markets absorb new demand far more slowly than urban ones, and because cash transfers appear to add to inflationary pressure precisely where markets are thinnest ([Cunha et al., 2019](#)), sustaining in-kind food assistance for longer after a displacement shock, or pairing cash transfers with measures that strengthen local market supply, may matter most precisely in the areas where aid coverage currently erodes fastest and cash transfers are already more common. Several limitations qualify how far these results travel beyond this setting. The analysis covers a single country over a period of unusually intense conflict, 2018–2022, and the three mechanisms identified here—market integration, aid coverage, and workforce composition—need not combine the same way in displacement driven by other causes, or in other regions. The one-year and eight-month windows used throughout also leave open how persistent these effects are: whether rural hosts remain pulled into agriculture as markets gradually catch up with demand, or whether the occupational upgrading observed in urban areas eventually reaches rural ones as well, is a question this design cannot answer, and one that further waves of the ESS, or comparable panel data from other displacement settings, could speak to directly.

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A Appendix

A.1 Model derivations and proofs

This appendix provides the derivations underlying the comparative statics in Section 3. All notation follows the main text.

A.1.1 Demand and goods-market adjustment

Cobb–Douglas preferences imply expenditure shares $\theta_F = \alpha$ and $\theta_N = 1 - \alpha$. Given fixed aggregate nominal host expenditure E_e^H and outside purchasing power b_e per displaced household, total

demand in sector s and location e is

$$Q_{se}^d(p_{se}, D) = \frac{\theta_s(E_e^H + b_e D)}{p_{se}}. \quad (44)$$

Consequently,

$$\frac{\partial Q_{se}^d}{\partial p_{se}} = -\frac{Q_{se}^d}{p_{se}}, \quad \kappa_{se} \equiv \frac{1}{Q_{se}^d} \frac{\partial Q_{se}^d}{\partial D} = \frac{b_e}{E_e^H + b_e D} \equiv \kappa_e. \quad (45)$$

Thus demand is unit elastic with respect to its own price and the proportional direct demand shift is identical across sectors within a location.

In a rural sector, market clearing requires $S_{sR}(p_{sR}, D) = Q_{sR}^d(p_{sR}, D)$. Totally differentiating with respect to D , dividing by equilibrium quantity, and applying the definitions of η_{sR} , λ_{sR} , and κ_R gives

$$\eta_{sR} \frac{d \log p_{sR}}{dD} + \lambda_{sR} = \kappa_R - \frac{d \log p_{sR}}{dD}. \quad (46)$$

Therefore,

$$\frac{d \log p_{sR}}{dD} = \frac{\kappa_R - \lambda_{sR}}{1 + \eta_{sR}} = \frac{\delta_{sR}}{1 + \eta_{sR}}. \quad (47)$$

In the urban location, $p_{FU} = p_F^*$, while services clear locally. Hence

$$\frac{d \log p_{FU}}{dD} = 0, \quad \frac{d \log p_{NU}}{dD} = \frac{\kappa_U - \lambda_{NU}}{1 + \eta_{NU}}. \quad (48)$$

Proof of Proposition 1. Using equation (47) in both rural sectors,

$$\frac{d}{dD} \log \left(\frac{p_{FR}}{p_{NR}} \right) = \frac{\kappa_R - \lambda_{FR}}{1 + \eta_{FR}} - \frac{\kappa_R - \lambda_{NR}}{1 + \eta_{NR}}. \quad (49)$$

Let $\Delta \lambda_R = \lambda_{FR} - \lambda_{NR}$. The right-hand side of equation (49) is positive if and only if

$$\frac{\kappa_R - \lambda_{NR} - \Delta \lambda_R}{1 + \eta_{FR}} > \frac{\kappa_R - \lambda_{NR}}{1 + \eta_{NR}}, \quad (50)$$

$$\Delta \lambda_R < \frac{(\kappa_R - \lambda_{NR})(\eta_{NR} - \eta_{FR})}{1 + \eta_{NR}}. \quad (51)$$

The second inequality is condition (iii) of Proposition 1. In the urban location, equation (48) and $\kappa_U > \lambda_{NU}$ imply

$$\frac{d}{dD} \log \left(\frac{p_{FU}}{p_{NU}} \right) = -\frac{\kappa_U - \lambda_{NU}}{1 + \eta_{NU}} < 0. \quad (52)$$

This proves both relative-price results.

Condition (iii) also implies positive net demand pressure in rural food. Since $\eta_{FR} < \eta_{NR}$,

$$0 < \frac{\eta_{NR} - \eta_{FR}}{1 + \eta_{NR}} < 1.$$

Together with $\kappa_R > \lambda_{NR}$, condition (iii) therefore gives

$$\lambda_{FR} - \lambda_{NR} < \kappa_R - \lambda_{NR}, \quad \delta_{FR} = \kappa_R - \lambda_{FR} > 0. \quad (53)$$

A.1.2 Worker optimization and occupational choice

Conditional on sector s , a type- z incumbent solves

$$\max_{c_F, c_N, h} \{C(c_F, c_N) - v(h)\} \quad \text{subject to} \quad p_F c_F + p_N c_N \leq a_s(z) w_s h. \quad (54)$$

Cobb–Douglas cost minimization yields the price index $P = p_F^\alpha p_N^{1-\alpha}$ and expenditure function PC . Since utility is increasing in consumption, the budget constraint binds and

$$C = \frac{a_s(z) w_s}{P} h. \quad (55)$$

Define the skill-adjusted real wage $r_s(z) = a_s(z) w_s / P$. The remaining problem is

$$\max_{h \geq 0} \left\{ r_s(z) h - \chi \frac{h^{1+1/\varepsilon}}{1+1/\varepsilon} \right\}. \quad (56)$$

The first-order condition and optimal hours are

$$r_s(z) = \chi h_s(z)^{1/\varepsilon}, \quad h_s(z) = \left(\frac{r_s(z)}{\chi} \right)^\varepsilon. \quad (57)$$

Using the first-order condition, maximized utility can be written as

$$U_s^*(z) = \frac{\chi^{-\varepsilon}}{1+\varepsilon} r_s(z)^{1+\varepsilon}. \quad (58)$$

It is strictly increasing in $a_s(z) w_s$. Given $a_F(z) = 1$ and $a_N(z) = z$, the worker chooses services if and only if

$$z w_{Ne} \geq w_{Fe} \quad \Longleftrightarrow \quad z \geq \bar{z}_e \equiv \frac{w_{Fe}}{w_{Ne}}. \quad (59)$$

The incumbent food-employment share is $G_e(\bar{z}_e)$. Its response to displacement is

$$\frac{dG_e(\bar{z}_e)}{dD} = g_e(\bar{z}_e) \bar{z}_e \left[\frac{d \log w_{Fe}}{dD} - \frac{d \log w_{Ne}}{dD} \right]. \quad (60)$$

A.1.3 Product prices and wages at a fixed cutoff

Let $r_{se} = w_{se}/P_e$, $q_{se} = p_{se}/P_e$, and $\gamma_s = 1 - \beta_s > 0$. Holding the assignment sets $\mathcal{Z}_{se}$ fixed, aggregate efficiency units of labor are

$$L_{se} = B_{se} r_{se}^\varepsilon, \quad B_{se} = \chi^{-\varepsilon} \int_{z \in \mathcal{Z}_{se}} a_s(z)^{1+\varepsilon} dG_e(z). \quad (61)$$

Thus $d \log L_{se} = \varepsilon d \log r_{se}$. Dividing the competitive labor demand condition by P_e gives

$$r_{se} = q_{se} A_s \beta_s L_{se}^{-\gamma_s}. \quad (62)$$

Log differentiation and substitution of labor supply yield

$$d \log r_{se} = \frac{1}{1 + \varepsilon \gamma_s} d \log q_{se}. \quad (63)$$

The price index implies the identities

$$d \log q_{Fe} = (1 - \alpha) d \log(p_{Fe}/p_{Ne}), \quad d \log q_{Ne} = -\alpha d \log(p_{Fe}/p_{Ne}). \quad (64)$$

These identities only express the product-price change in units of the local consumption basket. They contain no additional equilibrium restriction.

A.1.4 Moving cutoff and proof of Proposition 2

When the cutoff moves, the labor-supply shifters become

$$B_{Fe}(\bar{z}_e) = \chi^{-\varepsilon} G_e(\bar{z}_e), \quad B_{Ne}(\bar{z}_e) = \chi^{-\varepsilon} \int_{\bar{z}_e}^{\infty} z^{1+\varepsilon} dG_e(z). \quad (65)$$

Define

$$\mu_{Fe} \equiv \frac{\partial \log B_{Fe}}{\partial \log \bar{z}_e} = \frac{\bar{z}_e g_e(\bar{z}_e)}{G_e(\bar{z}_e)}, \quad \mu_{Ne} \equiv -\frac{\partial \log B_{Ne}}{\partial \log \bar{z}_e} = \frac{\bar{z}_e^{2+\varepsilon} g_e(\bar{z}_e)}{\int_{\bar{z}_e}^{\infty} z^{1+\varepsilon} dG_e(z)}. \quad (66)$$

Both terms are nonnegative. Since $d \log \bar{z}_e = d \log r_{Fe} - d \log r_{Ne}$, aggregate labor supply satisfies

$$d \log L_{Fe} = (\varepsilon + \mu_{Fe}) d \log r_{Fe} - \mu_{Fe} d \log r_{Ne}, \quad (67)$$

$$d \log L_{Ne} = -\mu_{Ne} d \log r_{Fe} + (\varepsilon + \mu_{Ne}) d \log r_{Ne}. \quad (68)$$

Let $x_s = d \log r_{se}$ and $q_s = d \log q_{se}$. Combining these labor-supply expressions with labor demand gives

$$\begin{pmatrix} 1 + \gamma_F(\varepsilon + \mu_{Fe}) & -\gamma_F \mu_{Fe} \\ -\gamma_N \mu_{Ne} & 1 + \gamma_N(\varepsilon + \mu_{Ne}) \end{pmatrix} \begin{pmatrix} x_F \\ x_N \end{pmatrix} = \begin{pmatrix} q_F \\ q_N \end{pmatrix}. \quad (69)$$

Its determinant is

$$\begin{aligned}\Delta_e &= (1 + \gamma_F \varepsilon)(1 + \gamma_N \varepsilon) \\ &\quad + \gamma_F \mu_{Fe}(1 + \gamma_N \varepsilon) + \gamma_N \mu_{Ne}(1 + \gamma_F \varepsilon) > 0.\end{aligned}\tag{70}$$

Writing $q_e = d \log(p_{Fe}/p_{Ne})$ and applying equation (64), the solution for the relative wage is

$$d \log \left(\frac{w_{Fe}}{w_{Ne}} \right) = \frac{1 + \varepsilon[(1 - \alpha)\gamma_N + \alpha\gamma_F]}{\Delta_e} d \log \left(\frac{p_{Fe}}{p_{Ne}} \right).\tag{71}$$

The coefficient is positive. Proposition 1 therefore implies $d\bar{z}_R/dD > 0$ and $d\bar{z}_U/dD < 0$. Equation (60) then implies that the incumbent food-employment share is nondecreasing in R and nonincreasing in U . The changes are strict whenever the density at the corresponding cutoff is positive. This proves Proposition 2.

Equation (71) also formalizes attenuation from occupational reallocation. Positive μ_{Fe} or μ_{Ne} increases Δ_e without changing the numerator, reducing relative-wage pass-through. If $g_e(\bar{z}_e) = 0$, both extensive-margin elasticities vanish and equation (63) is exact. If the density is sufficiently small, the fixed-cutoff signs of the individual real wages continue to hold by continuity. In particular, in the rural economy,

$$\frac{d \log h_F(z)}{dD} = \varepsilon \frac{d \log r_{FR}}{dD} > 0\tag{72}$$

when the density at the rural cutoff is zero or sufficiently small.

A.2 Construction of Price Variables

The price outcomes are constructed from the Ethiopia ESS community price module and household food purchase records. Let h index households, t survey years, i food items, $w(h)$ the woreda of household h , and $e(h)$ the enumeration area (EA) of household h . For each household-year, we first compute food expenditure shares from the household's reported food purchases. Let x_{hit} denote household h 's expenditure on food item i in year t , and let $\mathcal{I}_{ht}$ denote the set of food items purchased by household h that can be matched to community-survey price items. The household expenditure share for item i is

$$s_{hit} = \frac{x_{hit}}{\sum_{j \in \mathcal{I}_{ht}} x_{hjt}}.\tag{73}$$

These weights vary across households and years and capture the composition of the household's observed food purchase basket.

We construct two local food price deflators. For the woreda-based deflator, let $\bar{p}_{w(h)it}$ denote the average community-survey price of item i in household h 's woreda in year t , expressed in birr per kilogram, and let $\bar{p}_{it}^N$ denote the national item-year average price. The household-specific woreda

price deflator is

$$D_{ht}^W = \sum_{i \in \mathcal{I}_{ht}^W} s_{hit}^W \left(\frac{\bar{p}_{w(h)it}}{\bar{p}_{it}^N} \right), \quad (74)$$

where $\mathcal{I}_{ht}^W$ is the subset of purchased items with valid matched woreda-level community prices. The EA-based deflator is analogous, but uses average community-survey prices from the household's enumeration area:

$$D_{ht}^{EA} = \sum_{i \in \mathcal{I}_{ht}^{EA}} s_{hit}^{EA} \left(\frac{\bar{p}_{e(h)it}}{\bar{p}_{it}^N} \right). \quad (75)$$

Thus, both deflators aggregate item-level local-to-national price ratios using household food expenditure shares as weights. They differ only in the geographic level at which local community prices are measured: the first uses woreda-level average prices, while the second uses EA-level average prices.

We also construct a standardized market-price index. Let

$$z_{w(h)it} = \frac{\bar{p}_{w(h)it} - \mu_{it}}{\sigma_{it}}, \quad (76)$$

where μ_{it} and σ_{it} are the national mean and standard deviation, respectively, of community-survey prices for item i in year t . The household-level standardized price index is

$$Z_{ht} = \sum_{i \in \mathcal{I}_{ht}^Z} s_{hit}^Z z_{w(h)it}. \quad (77)$$

Unlike the deflators, this measure is expressed in standard-deviation units rather than local-to-national price-ratio units. A higher value indicates that the items in a household's food purchase basket are relatively more expensive in local community markets compared with the national item-year price distribution.

A.3 Food Security Indices

This appendix section discusses the issues due to the different time recalls in the food security sections in the ESS. The restricted index equalizes the content of the questions across waves but not their recall, and neither index escapes the change in window. Standardizing within wave removes any difference in level and scale between the two, and the time fixed effects of Equation 34 would absorb a common shift in any case. What remains is that the two windows do not capture the same object. A seven-day window records whether a household was food insecure at the moment of the

interview, which makes it sensitive to the timing of fieldwork within the agricultural calendar—June to August in wave 4, April and May in wave 5. A twelve-month window instead records whether any episode of insecurity occurred over the year, a coarser measure that is closer to saturation and less responsive to conditions on the day. Since the estimates identify from the change in a household’s index between the two waves, this change in measurement enters the estimate directly, and its consequences cannot be signed in advance. Results based on these measures are accordingly treated as suggestive rather than as estimates of an effect on a well-defined quantity of food security.

A.4 BHJ 2022 for the Agricultural Analysis Sample

The shift-level estimator applied to the agricultural analysis sample is a simplified version of the procedure set out in Section 5.4. Two features account for the simplification. First, the post-planting survey of the ESS is administered only in rural enumeration areas, so $U_a = 0$ throughout this sample and the urban interaction in Equation 34 drops out, leaving a single treatment coefficient β to be estimated. Second, every outcome drawn from the post-planting survey is recorded for the household as a whole, so no individual index is required. With one endogenous regressor and one instrument, the system is just identified and the GMM problem of Section 5.4 reduces to a two-stage least squares regression run at the level of the shocks.

The first step is to residualize the outcome and the treatment with respect to the controls and the fixed effects. For $v_{ht} \in \{y_{ht}, \Delta IDP_{at}\}$,

$$v_{ht}^\perp = v_{ht} - (\mathbf{X}_{ht}'\Pi + \lambda_h + \delta_t), \quad (78)$$

where ΔIDP_{at} is the eight-month flow of Equation 33 and $\mathbf{X}_{ht}$ collects the household controls described in Section 5.

The residualized variables are then aggregated to the origin-year level using the enclave shares. Since the shares are an attribute of the enumeration area, the aggregation factors into a sum over areas of an area-level share multiplying the residuals of the households sampled there:

$$\tilde{v}_{ot} = \sum_a s_{ao} \sum_{h \in a} v_{ht}^\perp. \quad (79)$$

Aggregating share-weighted sums rather than share-weighted means is what delivers the numerical equivalence described below: running the shift-level regression unweighted on these sums reproduces the weighted regression on origin-year means that [Borusyak et al. \(2022\)](#) derive, since the exposure weights are already embedded in $\tilde{v}_{ot}$.

Two-stage least squares is then run on the aggregated data, with the recentered origin-woreda conflict shock instrumenting for aggregated exposure. The first stage is

$$\widetilde{\Delta IDP}_{ot} = \rho c_{ot}^{8mths} + v_{ot}, \quad (80)$$

and the second stage is

$$\tilde{y}_{ot} = \beta \widehat{\Delta IDP}_{ot} + \tau_{ot}, \quad (81)$$

where c_{ot}^{8mths} counts conflict events in origin woreda o over the eight months preceding the post-planting interview, matching the window over which the treatment itself is measured.

The estimate of β obtained from Equation 81 is numerically identical to the one obtained by estimating Equation 34 at the household level, instrumenting the treatment with Z_{at}^{Ag} and imposing $U_a = 0$. What the shift-level formulation changes is the unit at which inference is conducted: because the observations of Equations 80 and 81 are indexed by origin and year, standard errors can be clustered at the woreda of origin, which is the level at which the identifying shocks vary.

Table 3: Summary Statistics for variables in the household analysis sample

Variable	Rural			Urban			Diff.
	N	Mean	SD	N	Mean	SD	
<i>Household annual consumption (household-year)</i>							
Total exp.	1,539	6,655.49	4,959.04	2,661	8,551.59	5,932.19	1,896.10***
Food exp.	1,539	5,348.63	4,155.01	2,661	5,225.38	3,770.16	-123.25
Deflated food exp.	1,486	5,692.96	4,821.85	2,583	5,803.57	4,779.48	110.61
Non-food exp.	1,539	743.56	669.10	2,661	1,548.02	1,428.52	804.46***
<i>Food price variables (household-year)</i>							
Woreda price deflator	1,486	1.00	0.24	2,583	0.97	0.21	-0.03***
EA price deflator	1,486	0.99	0.23	2,571	0.96	0.22	-0.03***
Price index (SD)	1,498	0.13	0.39	2,628	0.02	0.36	-0.11***
<i>Food security (household-year)</i>							
Full index (SD)	1,539	-0.08	1.06	2,661	0.09	0.96	0.17***
Full index (percentile)	1,539	47.98	27.21	2,661	52.63	25.37	4.65***
Common-items index (SD)	1,539	-0.07	1.04	2,661	0.08	0.96	0.14***
Common-items index (percentile)	1,539	48.39	26.34	2,661	51.97	24.74	3.58***
<i>Labor outcomes in last 7 days (individual-year, ages 15+)</i>							
Any IGA	4,538	0.49	0.50	7,423	0.41	0.49	-0.08***
Any ag work for HH	4,538	0.44	0.50	7,423	0.05	0.23	-0.38***
Any self-employment	4,538	0.04	0.19	7,423	0.12	0.33	0.08***
Any wage employment	4,538	0.03	0.16	7,423	0.22	0.41	0.19***
Total hours of labor	5,063	9.80	16.76	7,909	15.78	24.52	5.98***
Hours in ag work for HH	5,063	7.86	14.13	7,909	1.24	6.52	-6.63***
Hours in self-employment	5,063	0.77	5.31	7,909	4.08	13.17	3.31***
Hours in wage employment	5,063	0.72	5.46	7,909	8.28	17.88	7.56***
<i>Demographic variables (individual-year, ages 15+)</i>							
Household size	5,063	6.79	2.74	7,909	5.29	2.48	-1.50***
Age	5,063	33.02	15.82	7,909	33.83	14.94	0.82***
Male	5,063	0.50	0.50	7,909	0.44	0.50	-0.05***
<i>Education variables (individual-year, ages 15+)</i>							
Can read and write	4,538	0.44	0.50	7,423	0.82	0.38	0.38***
Ever attended school	4,538	0.46	0.50	7,423	0.81	0.39	0.35***
Currently attending school	4,537	0.19	0.39	7,422	0.22	0.41	0.03***
Completed primary school	4,489	0.14	0.35	7,334	0.59	0.49	0.45***
Completed high school	4,489	0.07	0.26	7,334	0.43	0.50	0.36***
<i>Conflict variables within 20 km (individual-year, ages 15+)</i>							
Conflict events	5,063	1.26	2.39	7,909	6.42	6.07	5.16***
Fatal conflict events	5,063	0.79	1.77	7,909	2.56	1.83	1.77***
Civilian-targeting events	5,063	0.84	1.45	7,909	2.46	1.97	1.62***

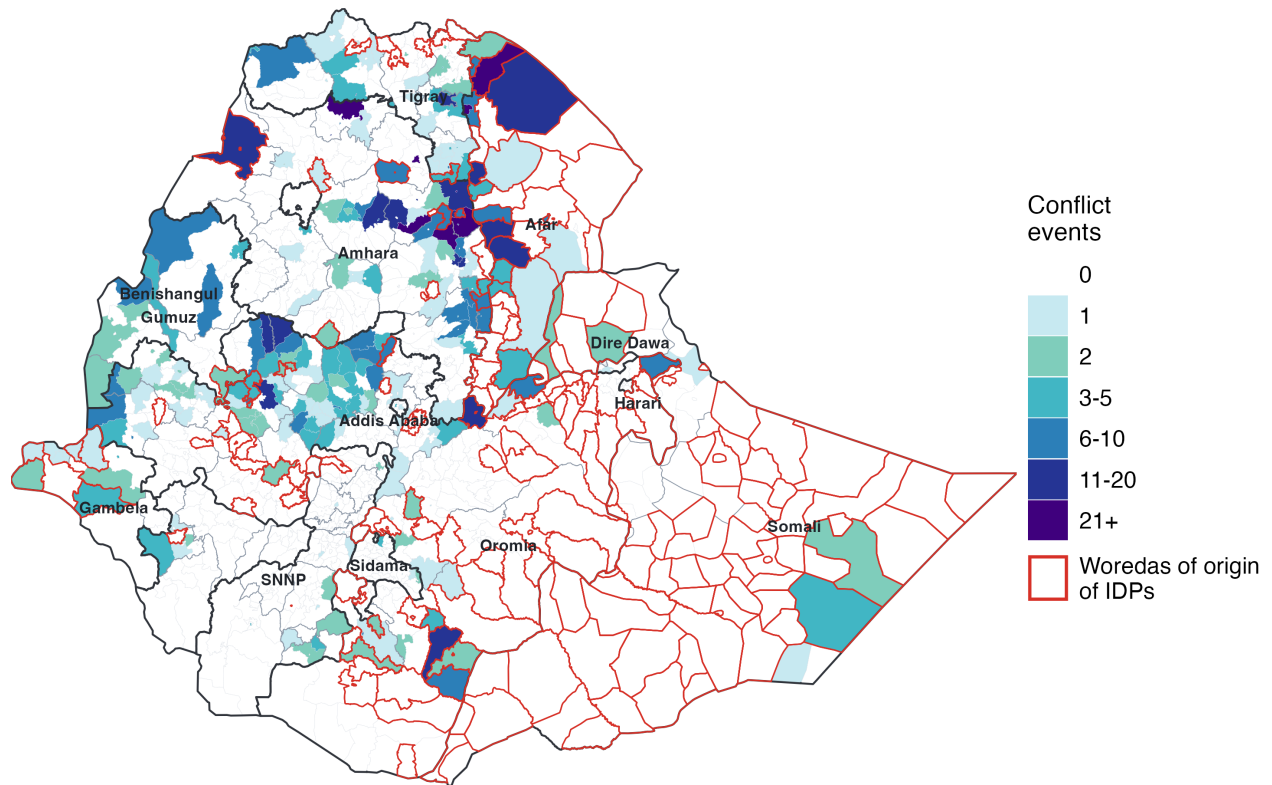
Notes: Table reports non-missing observations, means, and standard deviations for the final Ethiopia ESS household-survey analysis samples, separately for rural and urban households. All rows are restricted to households in DTM geographic coverage, exposed to IDPs in the R11 fixed-share baseline, and observed in both Waves 4 and 5. For the household annual consumption and price-variable sections, N is the number of household-year observations with non-missing values for the listed variable. For the labor outcomes and control variables sections, N is the number of adult individual-year observations with non-missing values for the listed variable, among adults aged 15 or older in the same household analysis sample. Annual total, food, local-price deflated food, and non-food expenditures are converted to 2021 PPP dollars using the CPI/PPP conversion used in the analysis scripts. The p-value column reports a Welch two-sample t-test of equality of rural and urban means.

Table 4: Summary Statistics for variables in the agricultural analysis sample

Variable	N	Mean	SD
<i>Agricultural outcomes (household-year)</i>			
Active crop cultivation	658	0.93	0.25
Hours of family labor for land preparation	590	301.25	337.91
Person-days of hired labor for land preparation	590	6.92	15.29
Daily wage for hired labor for land preparation	78	8.71	5.41
Any cultivation of staple crops	582	0.93	0.26
Any cultivation of cash crops	582	0.59	0.49
Potential rent/ha for ag land	374	2,252.33	2,324.20
<i>Demographic variables (individual-year, ages 15+)</i>			
Household size	3,324	6.62	2.76
Age	3,324	33.09	15.65
Male	3,324	0.50	0.50
<i>Education variables (individual-year, ages 15+)</i>			
Can read and write	2,904	0.46	0.50
Ever attended school	2,904	0.50	0.50
Currently attending school	2,904	0.18	0.39
Completed primary school	2,867	0.16	0.37
<i>Conflict variables within 20 km (household-year)</i>			
Conflict events	658	2.06	2.80
Fatal conflict events	658	1.19	1.98
Civilian-targeting events	658	1.46	1.71

Notes: Table reports non-missing observations, means, and standard deviations for the rural Ethiopia ESS agricultural analysis sample. The sample is restricted to households in DTM geographic coverage, exposed to IDPs in the R9 fixed-share baseline, and observed in both Waves 4 and 5. For the agricultural outcomes section, family-labor and hired-labor rows are restricted to model-complete post-planting household-year observations that are active in crop cultivation. Active crop cultivation is defined as reporting household crop activity, any cultivated plot, or a positive number of cultivated plots. Cultivation outcomes are post-harvest indicators; labor and land outcomes use the post-planting samples. Standard deviations for binary cultivation outcomes are computed from the analysis-sample mean and N; standard deviations for labor and land-rent outcomes are computed from the corresponding raw model-complete analysis samples. For the demographic, education, and conflict-variable sections, N is the number of adult individual-year observations with non-missing values for the listed variable, among adults aged 15 or older in the same rural post-planting household analysis sample. Hired-labor daily wages and land rent per hectare are converted to 2021 PPP dollars. Land rent per hectare uses plots with positive potential rent and GPS-measured plot area at least 0.05 ha, with GPS area converted from square meters to hectares, and winsorizes household-year means at the 5th and 95th percentiles.

Figure 4: Conflict events and Woredas of origin of IDPs for 2019



Notes: ACLED conflict events are counted over 16 Apr 2021 to 15 Apr 2022, matching the 2022 rural/urban shift window. The blue fill shows the raw count of conflict events in each woreda. Zero-event woredas are white. Colors are binned to distinguish low-conflict woredas: 1 event, 2 events, and 3-5 events are shown separately. Red outlines mark R11 host-community/camp IDP origin woredas used in the enclave shares. This is the raw shift underlying the shift-share instrument. The regression instrument uses the recentered shift, weighted by R11 enclave shares. Boundaries show Ethiopia ADM3 woredas, ADM2 zones, and ADM1 regions, from thinner to thicker outlines.

Table 5: Summary Statistics for Conflict and share support in Woredas of origin of IDPs

Variable	N	Mean	SD	Min	Max
Conflict events in origin woreda, prior year	144	0.83	2.88	0.00	21.00
Household-years with positive share	144	36.55	128.38	1.00	1289.00
Mean household origin share	144	0.62	0.36	0.02	1.00
Mean baseline IDP stock from origin woreda	144	2440.71	2751.56	81.00	14789.00

Notes: Statistics describe the origin-woreda data used to construct the Ethiopia R11 enclave instrument for host communities and camps. The instrument fixes destination-origin shares using DTM round R11 and links ESS household-years in 2019 and 2022 to DTM destinations within 20 km. Conflict variables are counted in the origin woreda over the 365 days before each ESS interview reference date: June 15, 2019 for wave 4 and April 15, 2022 for wave 5. The table uses one observation per origin woreda by ESS survey year among origin woredas with a positive share for at least one household-year. Mean baseline IDP stock is the average number of baseline IDPs from that origin woreda across linked household-year observations. The conflict-event count is based on all ACLED events matched by admin3/origin-woreda name.

Table 6: First Stage Statistics

	IDP flow within 20km $\times$ rural household	IDP flow within 20km $\times$ urban household
Recentered shock	0.409*** (0.026)	0.298*** (0.042)
BH 2023 RI p-value	0.017	0.030
Sanderson-Windmeijer F	217.99	41.34
SW F p-value	2.47×10^{-23}	1.28×10^{-8}
SW F degrees of freedom	1, 71	1, 71
Kleibergen-Paap rk Wald F		60.01
KP rk Wald F p-value		4.94×10^{-11}
Kleibergen-Paap rk Wald χ^2		120.01
KP degrees of freedom		1
Origin-year cells	104	143
Effective N	11.60	8.76
Origin-woreda clusters	52	72

Notes: The table reports BHJ 2022 shift-level first-stage regressions and diagnostics for the total annual consumption rural-urban interaction specification. The shift-level system stacks the base R11-share moment and the rural-household R11-share moment. Household-level outcome and treatment variables are first residualized on household fixed effects, year fixed effects, household demographic controls, education controls, and local conflict controls; the residualized variables are then aggregated to the shift level using the BHJ exposure weights. Columns report the interacted specification in household-group terms: the rural column corresponds to the 1-year IDP flow within 20km, scaled in 1,000 IDPs, interacted with the rural household indicator; the urban column corresponds to the base treatment term, i.e. the treatment slope for urban households. The coefficient row reports the relevant recentered shift-level shock: the rural recentered shock for the rural-interaction first stage and the base recentered shock for the urban first stage. Standard errors are in parentheses and are clustered by woreda of origin. Significance levels: * p $\leq$ 0.10, ** p $\leq$ 0.05, *** p $\leq$ 0.01. The middle rows report Sanderson-Windmeijer and Kleibergen-Paap first-stage diagnostics computed on the same weighted shift-level stack. Effective N is defined as the inverse Herfindahl index, 1 divided by the sum of squared normalized shift-cell weights, computed separately within each column. The stacked shift-level sample has 247 rows, 72 origin-woreda clusters, 143 base shift cells, and 104 rural-interaction shift cells.